

Extra Capstone Problems

Simulation Without a Black Box: Modeling and Simulation in the AI Era

Version 1.0

These 27 problems are open-ended capstone studies drawn from real operational settings across healthcare, manufacturing, logistics, transportation, computing, and other domains. Each problem specifies a system, representative data, key performance indicators, and a structured set of deliverables. Every problem is written at three levels of engagement; students and instructors choose the level appropriate to their course objective.

Three learning tracks per problem:

Track A — Industrial Case Study. Standard deliverables only. Build, validate, and analyze the simulation model; report findings as a consulting memo for the specific system described. Appropriate for course projects and industry practice. All 27 problems support Track A.

Track B — Research Study. Add the *Research Literature* and *Research Track Deliverables* sections. Before building the model, conduct a literature review of state-of-the-art DES and analytical work on the domain. Pose a falsifiable research hypothesis; design a factorial experiment to test it; compare simulation results to an analytical approximation; and frame generalizable conclusions. The output is a 6–10 page report suitable for submission to the *Winter Simulation Conference* (WSC) or a domain journal. Problems EP01, EP06, EP13, EP20, EP21, EP24, and EP25 have full Research Track Deliverables; the Research Literature section is provided for all problems.

Track C — Simulation as Environment (RL). Add the *Advanced RL Extension* section. Formulate the management decision as a Markov Decision Process (MDP); implement the SimPy simulation as a `Gymnasium Env`; train a reinforcement learning agent (e.g., PPO, DQN, or SAC); and compare the learned policy against the best rule-based heuristic. This track is publishable at WSC, INFORMS, or domain-specific ML/OR venues as a *simulation-as-environment* research contribution. The core insight: the transparent SimPy model enables interpretable policy comparison and counterfactual analysis that black-box COTS simulators cannot provide. All 27 problems support Track C.

Data policy. The numerical parameters in each problem are *representative estimates* that let you build and test a model without fieldwork. In a real consulting engagement, you would replace these figures with primary data collected through observation, electronic records, or published benchmarks. Problems note which parameters are most sensitive to measurement error and suggest data-collection approaches.

Solutions. Instructor solutions (LLM prompts, Python code, output analysis, and a model report) are distributed to instructors only, as part of the instructor courseware package.

Contents

1 Healthcare Systems

EP01 — Emergency Department Patient Flow

Domain: Healthcare • **Difficulty:** *** • **Key techniques:** Non-stationary arrivals, priority queuing, resource contention, patient boarding

System Overview. An urban academic emergency department (ED) operates 24 hours a day, seven days a week. Patients arrive by ambulance or walk-in; acuity is classified by a triage nurse using the five-level Emergency Severity Index (ESI). High-acuity patients may preempt lower-acuity patients for treatment rooms and physician attention. Patients who require inpatient admission wait in the ED for an inpatient bed (boarding), which drives crowding.

Operational Description. On arrival a patient waits for triage (single queue, two triage nurses), receives an ESI level, and then waits for a treatment room. The ED has 30 treatment rooms shared across all acuity levels; ESI 1–2 patients have priority. Once in a room, patients may require laboratory tests or imaging; results take additional time and hold the room. After the physician completes the encounter, patients are discharged, admitted to an inpatient unit, or transferred. Admitted patients board in the ED until an inpatient bed is allocated (exponentially distributed wait). Patients who have not seen a provider within 90 minutes may leave without being seen (LWBS).

Data.

Parameter	Value
Arrival rate (hourly, weekday)	2/hr (00–05), 5/hr (06–09), 9/hr (10–14), 11/hr (15–19), 7/hr (20–23)
Weekend multiplier	+25% across all hours
ESI distribution	L1 2%, L2 8%, L3 35%, L4 40%, L5 15%
Triage time	Exp(4 min)
Treatment time (L1)	LN(180 min, 60 min)
Treatment time (L2–3)	LN(135 min, 50 min)
Treatment time (L4–5)	LN(50 min, 20 min)
Lab/imaging request prob.	L1–2: 90%, L3: 75%, L4: 40%, L5: 10%
Lab turnaround	LN(65 min, 30 min)
Imaging turnaround	LN(45 min, 20 min)
Admission rate	L1–2: 80%, L3: 35%, L4: 5%, L5: 0%
Inpatient boarding time	Exp(90 min)
Physicians	4 (day 07–19), 3 (evening 15–23), 2 (night 23–07)
Triage nurses	2 at all times
Treatment rooms	30
LWBS patience	90 min without provider contact

Arrival rates and ESI distributions are representative of a Level I trauma center in a mid-sized US city. Primary sources for calibration include the hospital's ADT system and triage logs. LWBS patience thresholds vary by institution; validate against patient-experience surveys.

Key Performance Indicators.

- Door-to-triage time (median and 90th percentile); target ≤ 10 min
- Door-to-provider time (median and 90th percentile); target ≤ 30 min
- Length of stay by ESI level
- LWBS rate; target $< 2\%$
- Boarding time for admitted patients
- ED occupancy (beds occupied / 30) throughout the day

Modeling Scope. Model triage, treatment room assignment, physician and nurse resource contention, lab/imaging delays, and inpatient boarding. Exclude surgical suite scheduling and inpatient unit dynamics beyond the boarding wait distribution.

Assumptions.

- Physician and nurse counts follow the three-shift schedule; no float pool.
- Lab and imaging departments have infinite capacity; only turnaround time is modeled.
- A patient requires exactly one physician and one room simultaneously.
- Ambulance arrivals and walk-ins are pooled into a single arrival process for each hour.
- Patients do not arrive during triage; the triage queue is served FCFS with priority pre-emption beginning at room assignment.

Deliverables.

1. Build and validate the baseline model; compare simulated daily throughput and mean LOS by acuity to published ED benchmarks.
2. Analyze door-to-provider time by hour of day; identify the two worst consecutive hours.
3. Evaluate a fast-track protocol: divert all ESI 4–5 patients to a dedicated 4-bed fast-track unit staffed by one physician assistant (PA); service time reduced to $LN(30, 12)$ min.
4. Staffing sensitivity: vary overnight physician count (± 1); report LWBS rate and average boarding time with 95% confidence intervals.
5. Boarding impact study: vary mean inpatient boarding time from 60 to 180 min in steps of 30 min; plot LWBS rate and ED occupancy.
6. Run 50 replications for each scenario; report paired confidence intervals (CRN) vs. baseline for each KPI.
7. Prepare a one-page executive summary with a staffing recommendation justified by simulation output, including uncertainty quantification.

Extensions.

1. Add a diversion policy: when ED occupancy exceeds 90%, ambulance arrivals are redirected to a neighboring hospital; model the probability that a diverted patient deteriorates and returns.
2. Incorporate psychiatric boarding as a separate patient subgroup (5% of admissions) with a specialized inpatient bed hold of $Exp(240 \text{ min})$; evaluate its disproportionate impact on overall crowding.

Research Literature. Search the *Winter Simulation Conference* (WSC) proceedings and *Health Care Management Science* for “emergency department simulation”, “LWBS crowding DES”, and “physician staffing boarding”. Before building the model,

understand the state of the art in non-stationary ED arrival modeling and the role of inpatient boarding in crowding. Your literature review should identify whether prior work has jointly studied physician shift design and boarding duration as interacting levers for LWBS — most papers treat them separately.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: physicians free, rooms occupied (by ESI class), queue lengths per ESI level, hour of day, day of week.

Action $\mathcal{A}$: physician headcount for the next 8-hour shift $\{2, 3, 4, 5, 6\}$.

Decision epochs: shift handovers (every 8 hours); semi-MDP.

Reward r_t : $-\text{LWBS_count} - \alpha \cdot \text{DTP_p90} - \beta \cdot \text{staffing_cost}$ per shift period.

RL deliverables.

1. Implement the simulation as a `gymnasium.Env`; define observation and action spaces; verify with a random policy.
2. Train a PPO agent; compare its learned policy to the current fixed-schedule heuristic on LWBS rate and total staffing cost using 95 % CIs over 50 test episodes.
3. Visualize the agent's policy: for each (hour, occupancy) combination, plot the recommended staffing level; identify when the agent diverges from the clinical rule-of-thumb.
4. Ablation: remove the hour-of-day feature from the state; measure performance degradation — this quantifies how much of the agent's advantage comes from non-stationarity awareness.

Research Track Deliverables (Track B).

1. **Research hypothesis.** State a falsifiable claim, e.g., “adding one overnight physician reduces LWBS by more than boarding-time reduction from 90 to 60 min, for an ED operating at $\rho > 0.80$.”
2. **Analytical benchmark.** Derive an $M(t)/D/c/K$ approximation for the physician queue; compute the expected LWBS rate and compare to the simulation; document and explain every discrepancy > 0.3 pp.
3. **Factorial experiment.** Design a 3×3 full factorial: overnight physician count $\{2, 3, 4\} \times$ mean boarding duration $\{60, 90, 120\}$ min; 50 replications per cell; report LWBS rate with 95 % CIs in a response table.
4. **Regression metamodel.** Fit a second-order response surface to the 9 factorial cells; identify the main effects, interaction, and the region of near-optimal operating conditions.
5. **Generalizable conclusions.** Frame findings as a rule: “For EDs with boarding

mean $\geq X$ and overnight utilization $\geq Y$, staffing dominates boarding reduction by a factor of ...”

EP02 — Outpatient Clinic No-Shows and Overbooking

Domain: Healthcare • **Difficulty:** ** • **Key techniques:** Appointment scheduling, no-shows, overbooking, revenue modeling

System Overview. A primary-care clinic operates Monday through Friday, 08:00–17:00, with three physicians each seeing patients in 20-minute appointment slots. High no-show rates erode physician productivity and revenue; overbooking recovers capacity but risks excessive wait times when too many patients actually appear.

Operational Description. Each physician has 25 scheduled slots per day (9:00–17:00 with two 10-minute breaks absorbed into slot buffers), for 75 daily appointments. Patients are classified as new (higher no-show rate) or established (lower rate). The clinic applies a double-booking policy to patients with two or more prior no-shows. Walk-in patients are accepted if the estimated remaining wait is below a threshold; otherwise they are turned away. Service times follow a right-skewed distribution reflecting brief and complex encounters.

Data.

Parameter	Value
Scheduled slots/day	75 (25 per physician)
Slot duration	20 min
Patient mix	40% new, 60% established
No-show rate (new)	28%
No-show rate (established)	18%
Walk-in demand	Pois(10)/day
Walk-in acceptance threshold	estimated wait $\leq$ 45 min
Service time	Triangular(8, 12, 35) min
Double-booking candidates	20% of scheduled patients ($\geq$ 2 prior no-shows)
Physician overtime tolerance	up to 20 min past 17:00
Revenue per completed visit	\$180

No-show rates vary widely by specialty (8–30%) and patient population. Calibrate from the clinic's scheduling system. Service-time parameters should be measured by time-and-motion study for at least 100 encounters.

Key Performance Indicators.

- Physician utilization (time with patient / scheduled session time)
- Patient wait time from scheduled arrival to room entry
- Daily overflow: walk-in patients turned away
- No-show revenue loss (\$ per day)
- Overtime frequency (fraction of days any physician works past 17:00)

Modeling Scope. Model the daily clinic session for one physician and then scale to three. Capture appointment arrivals, no-show realizations, walk-in arrivals, double-booking collisions, and service-time variability. Exclude billing, referral pathways, and lab results that require a follow-up visit.

Assumptions.

- Patients arrive exactly at their scheduled time if they show up.
- Walk-in patients who are turned away do not retry the same day.
- Each physician operates an independent queue; no patient sharing across physicians.
- The overtime cap is hard: any patient not started by 16:50 is rescheduled.

Deliverables.

1. Simulate 240 clinic days at baseline; report mean and 95% CI for all KPIs.
2. Compare three no-show interventions: (a) automated phone reminder (reduces no-show rate by 8 percentage points), (b) SMS reminder (reduces by 12 pp), (c) 15% overbooking applied uniformly.
3. Optimize the double-booking fraction (0%, 10%, 15%, 20%) using a simulation search; report the Pareto frontier of average wait time versus daily revenue.
4. Find the maximum walk-in acceptance rate before physician utilization exceeds 95%.
5. Use common random numbers to produce paired confidence intervals for each intervention versus baseline on wait time and revenue.

Extensions.

1. Introduce same-day cancellations (8% of appointments, arriving uniformly throughout the morning); model the clinic’s ability to offer reclaimed slots to the walk-in waitlist.
2. Model provider-specific no-show rates and assess whether unequal overbooking across providers reduces total overtime.

Research Literature. Search WSC, *Health Care Management Science*, and *IIE Transactions on Healthcare Systems Engineering* for “outpatient clinic overbooking simulation”, “no-show prediction appointment scheduling DES”, and “advanced access scheduling”. Identify the dominant no-show modeling approaches (Bernoulli, heterogeneous patience) and understand what has been studied versus what remains open for your specific patient population.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: day type (Mon–Fri), predicted no-show rate (rolling 4-week average), confirmed appointments scheduled, current wait-list depth.

Action $\mathcal{A}$: overbooking level per slot $\{0, 1, 2, 3\}$ extra patients beyond nominal capacity.

Decision epochs: nightly (one decision per upcoming appointment day).

Reward r_t : $-\text{idle_physician_time} - \gamma \cdot \text{overtime} - \delta \cdot \text{patient_wait} + \epsilon \cdot \text{patients_seen}$.

RL deliverables.

1. Implement as a `gymnasium.Env`; test with a constant-overbooking baseline to confirm reward alignment.
2. Train a DQN agent; compare to the current fixed-overbooking policy on physician idle time, overtime, and no-show waste.
3. Evaluate generalization: test the trained agent when no-show rates shift by ± 5 pp from the training distribution; report performance degradation.

EP03 — ICU Surge Capacity Planning

Domain: Healthcare • **Difficulty:** ★★★ • **Key techniques:** Finite-capacity queues, diversion, surge modeling, step-down units

System Overview. A 20-bed medical ICU (MICU) operates continuously, admitting critically ill patients from the emergency department, operating suites, and direct

transfers. When the ICU is near capacity, new patients are diverted to a partner institution. A step-down unit provides an overflow buffer for patients who have partially recovered.

Operational Description. Patients arrive as a Poisson process; three clinical types (medical, surgical, trauma) have different length-of-stay distributions and ventilator requirements. When occupancy reaches or exceeds 18 of 20 beds, new arrivals are diverted; diversion increases a patient’s risk of deterioration. Stabilized ICU patients may be transferred to an 8-bed step-down unit; step-down patients may be re-admitted to the ICU with a given daily probability.

Data.

Parameter	Value
ICU beds	20
Step-down beds	8
Arrival rate (base)	Pois(2.8)/day
Arrival rate (surge, 7-day period)	Pois(5.6)/day
Patient mix	70% medical, 20% surgical, 10% trauma
LOS: medical	LN(3.5 days, 2.0 days)
LOS: surgical	LN(4.8 days, 2.5 days)
LOS: trauma	LN(5.1 days, 2.8 days)
Ventilator requirement	Medical 65%, surgical/trauma 80%
Ventilators available	12
Diversion threshold	≥ 18 beds occupied
Deterioration risk if diverted	40% of diverted patients experience delay harm
Step-down transfer probability	0.60 after $1.5 \times$ ICU LOS elapsed
Step-down re-admission rate	0.15/patient/day

ICU occupancy benchmarks: 70–85% occupancy is typical for US academic MICUs. Ventilator prevalence and LOS distributions should be calibrated from the hospital’s clinical data warehouse.

Key Performance Indicators.

- Mean and 95th-percentile ICU bed occupancy
- Diversion rate (% of arrivals diverted)
- Time from ICU request to bed assignment
- Ventilator utilization
- Step-down unit occupancy

Deliverables.

1. Simulate 365 base days; validate occupancy distribution against the 70–85% benchmark.
2. Inject a 7-day surge event; report diversion rate and expected number of harm events.
3. Capacity expansion: compare +4 beds, +8 beds, and +2 step-down beds; select the configuration that eliminates diversion at $\leq$ \$5M annualized cost (\$500K/bed/year).
4. Flexible staffing: activate a float-pool nurse when occupancy $\geq$ 16 beds; evaluate effect on care hours per patient day.
5. Report 95% CIs for diversion rate with and without flex staff across 30 replications.

Extensions.

1. Model step-down re-admission as a feedback loop; show how ICU and step-down interact under surge conditions.
2. Add a shared ventilator pool with a neighboring ICU (10 additional vents available with 2-hour request delay); evaluate ventilator shortage frequency.

Research Literature. Search WSC, *Critical Care Medicine*, and *Medical Care* for “ICU surge capacity simulation”, “hospital capacity planning DES”, and “diversion policy critical care”. Note how prior work models downstream step-down capacity and whether patient-mixing (elective deferral) has been jointly optimized with surge-bed activation.

Advanced RL Extension (Track C). MDP formulation.

State S : ICU beds occupied, step-down beds occupied, pending transfer queue length, elective-admission count today, surge-bed activation status $\{0, 1\}$.

Action A : surge beds to activate $\{0, 4, 8, 12\}$ and diversion status $\{0, 1\}$ (combined discrete action space).

Decision epochs: every 4 hours.

Reward r_t : $-\text{refusals} - \alpha \cdot \text{overflow_transfers} + \beta \cdot \text{ICU_utilization} - \gamma \cdot \text{surge_activation_cost}$.

RL deliverables.

1. Train PPO; compare to a threshold-based policy (activate surge when ICU occupancy $\geq$ 90%) on refusal rate and cost with 95 % CIs.
2. Test under a simulated pandemic surge (arrival rate $\times 1.5$ for 30 days); report policy robustness and worst-case refusal count.

EP04 — Regional Mass Vaccination Campaign

Domain: Healthcare • **Difficulty:** ★★ • **Key techniques:** Multi-stage service, appointment/walk-in mix, supply-chain integration

System Overview. A county public-health department operates five vaccination sites (school gymnasiums) for a 60-day campaign targeting 100,000 residents. Throughput, cycle time, vaccine waste, and campaign completion rate are the critical performance measures.

Operational Description. Each site runs 08:00–18:00 daily. Arrivals consist of scheduled appointments (pre-booked online) and walk-ins. Upon arrival, a patient passes through four sequential stages: registration, health screening, vaccination injection, and a mandatory 15-minute observation period. Vaccines are replenished every three days; unused doses expire after six days of preparation (thaw). An ordering trigger should be evaluated.

Data.

Parameter	Value
Sites	5
Campaign duration	60 days
Appointment slots/site/day	300
Walk-in demand	Pois(30)/site/day
Appointment no-show rate	15%
Registration time	Exp(2 min)
Health screening time	Exp(4 min)
Vaccination time	Exp(3 min)
Observation time	15 min (deterministic)
Registrars per site	2
Screeners per site	2
Vaccinators per site	4
Vaccine replenishment cycle	every 3 days; 320 doses/site/cycle
Dose shelf-life after thaw	6 days
Ordering lead time	2 days

Walk-in demand and no-show rates should be estimated from pilot-day observation and registration system data. Cold-chain parameters come from the vaccine manufacturer's product insert.

Key Performance Indicators.

- Daily throughput (vaccinations per site)
- Mean cycle time (door to exit)
- Vaccine waste rate (% of received doses expired unused)
- Stage-level wait time and utilization
- Campaign coverage: estimated day on which 100,000 doses administered

Deliverables.

1. Build a single-site model; validate throughput against theoretical maximum (limited by vaccination stage capacity).
2. Find the staff mix (registrars, screeners, vaccinators) that achieves ≥ 280 vaccinations/site/day with mean cycle time ≤ 25 min.
3. Simulate a 2-day vaccine supply disruption; report throughput loss and recovery time.
4. Vary walk-in demand (10, 30, 50/site/day); report wait-time distribution at each level.
5. Estimate the earliest day to reach 100,000 total doses under the optimal staffing configuration.

Extensions.

1. Model a cold-chain breakdown at one site (1% daily probability; site offline 4 hours); evaluate cross-site capacity reallocation.
2. Introduce geographic demand heterogeneity: sites in lower-income areas have 30% higher walk-in rates; evaluate equity of cycle-time outcomes across sites.

Research Literature. Search WSC, *Production and Operations Management*, and *Vaccine* for “mass vaccination campaign simulation”, “pandemic resource allocation DES”, and “vaccination station staffing”. Identify whether prior work has addressed the joint optimization of station count and inventory replenishment under demand uncertainty.

Advanced RL Extension (Track C). MDP formulation.

State S : vaccine inventory on hand, stations currently open, queue length, forecast demand for the next 2 hours, fraction of daily target achieved.

Action $\mathcal{A}$: number of active vaccination stations $\{1, 2, \dots, 10\}$.

Decision epochs: hourly.

Reward r_t : $-\text{mean_wait_per_patient} - \alpha \cdot \text{idle_staff_cost} - \beta \cdot \text{vaccine_waste_units}$.

RL deliverables.

1. Train DQN; compare to a fixed-station-count policy on throughput, waste, and total cost.
2. Vary demand forecast accuracy; show how RL performance degrades as forecast noise increases.

2 Manufacturing Systems

EP05 — Job Shop with Machine Breakdowns

Domain: Manufacturing • **Difficulty:** ★★★ • **Key techniques:** Job shop routing, dispatching rules, machine breakdowns, preventive maintenance

System Overview. A six-machine job shop fabricates five part types, each following a distinct routing through a subset of machines. Machine breakdowns are random and preempt current jobs. The study objective is to identify the dispatching rule and preventive-maintenance schedule that best balances throughput and on-time delivery.

Operational Description. Jobs arrive as a Poisson process and are assigned a part type upon arrival. Each part type has a fixed routing (sequence of machines) with lognormally-distributed processing times at each machine. When a machine breaks down, the job in process is interrupted and resumes after repair; waiting jobs remain in queue. Jobs receive due dates at arrival equal to their arrival time plus a multiple of the total expected processing time.

Data.

Processing times (mean $\pm$ std, minutes; “—” = not visited):

Machine	Part A	Part B	Part C	Part D	Part E
M1	18 ± 6	—	22 ± 8	—	15 ± 5
M2	—	25 ± 9	30 ± 10	—	—
M3	35 ± 12	—	—	28 ± 10	—
M4	—	20 ± 7	25 ± 9	40 ± 14	—
M5	22 ± 8	—	18 ± 6	—	32 ± 11
M6	15 ± 5	18 ± 6	—	22 ± 8	—

Routings: A: M1→M3→M5→M6; B: M2→M4→M6; C: M1→M2→M4→M5;
D: M3→M4→M6; E: M1→M5.

Parameter	Value
Job inter-arrival time	Exp(8 min)
Part type distribution	20% A, 25% B, 20% C, 20% D, 15% E
Processing time distribution	Lognormal (parameters above)
Machine MTTF	Exp(480 min) (all machines)
Machine MTTR	Exp(60 min) (all machines)
Due date factor	arrival + 3 × sum of mean processing times on route

Processing times and breakdown parameters should be estimated from production records and maintenance logs. MTTF/MTTR can be fit with exponential or Weibull distributions depending on failure mode.

Key Performance Indicators.

- Mean flow time (arrival to completion)
- Mean tardiness ($\max(0, \text{completion} - \text{due date})$)
- On-time delivery rate (%)
- Machine utilization per machine
- Mean WIP (jobs in system)

Deliverables.

1. Baseline with FCFS dispatching: validate machine utilizations against utilization bounds computed from arrival rate and mean processing times.
2. Conduct warm-up analysis; determine the minimum warm-up period for each KPI using Welch's method.
3. Compare three dispatching rules: FCFS, Shortest Processing Time (SPT), Earliest Due Date (EDD); report all KPIs with 95% CIs.

4. Breakdown sensitivity: reduce MTTF from 480 to 240 min; report change in mean tardiness and on-time rate.
5. Preventive maintenance policy: schedule a 30-min PM when each machine has consumed 50% of its expected MTTF since last repair; compare OTD to the no-PM baseline.

Extensions.

1. Add sequence-dependent setup times (20 min changeover when consecutive jobs at a machine differ in part type); re-evaluate all dispatching rules.
2. Model unplanned worker absence: one machine is unavailable for an entire 8-hour shift on a randomly chosen day each week; assess tardiness impact.

Research Literature. Search WSC and *International Journal of Production Research* for “dynamic scheduling simulation job shop”, “dispatching rules comparison DES”, and “predictive maintenance scheduling simulation”. Understand the standard comparison methodology for dispatching rules (SPT, EDD, ATC) and identify whether combined dispatching-and-maintenance policies have been studied for your machine configuration.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: for each machine: status (idle/busy/down), queue length, age since last PM;
global: WIP count, fraction of jobs past due.

Action $\mathcal{A}$: at each machine-free event, select next job from the queue and decide PM now vs. defer.

Decision epochs: each time a machine becomes idle.

Reward r_t : $-\text{tardiness} - \alpha \cdot \text{flow_time} - \beta \cdot \text{breakdown_count}$.

RL deliverables.

1. Train PPO to jointly dispatch and schedule PM; compare to the best fixed-rule combination (SPT + threshold PM) on mean tardiness and on-time delivery rate.
 2. Analyze the learned dispatching policy: at what machine age and WIP level does the agent switch from SPT-like to EDD-like behavior?
-

EP06 — Semiconductor Wafer Fabrication Re-entrant Flow

Domain: Manufacturing • **Difficulty:** *** • **Key techniques:** Re-entrant queues, bottleneck analysis, WIP control, CONWIP policy

System Overview. A 200 mm wafer fabrication facility produces logic chips through a seven-step process. The lithography step (step 3) is re-entered twice and constitutes the bottleneck. The study evaluates lot dispatching rules, hot-lot policies, and WIP-release controls to minimize cycle time without sacrificing throughput.

Operational Description. Wafer lots (25 wafers each) are released into the fab at a controlled rate and flow through processing steps in sequence. On the second visit to lithography, lots compete with newly arrived lots at the bottleneck queue. A fraction of lots are designated “hot lots” and preempt normal lots at lithography only. Equipment is subject to random breakdowns.

Data.

Step	Equipment (tools)	Processing time
1 Diffusion	3 tools	$\mathcal{N}(120 \text{ min}, 12 \text{ min})$
2 Photo	2 tools	$\mathcal{N}(45 \text{ min}, 5 \text{ min})$
3 Lithography	2 tools	$\mathcal{N}(95 \text{ min}, 8 \text{ min})$
4 Etch	3 tools	$\mathcal{N}(60 \text{ min}, 6 \text{ min})$
3* Lithography (2nd visit)	same 2 tools	$\mathcal{N}(85 \text{ min}, 7 \text{ min})$
5 CVD	2 tools	$\mathcal{N}(75 \text{ min}, 8 \text{ min})$
6 Test	2 tools	$\mathcal{N}(40 \text{ min}, 4 \text{ min})$

Parameter	Value
Lot release rate	10 lots/day (Exp(144 min) inter-release)
Hot-lot fraction	10% of lots
Lithography MTTF	Exp(2000 min)
Lithography MTTR	Exp(120 min)
Other equipment MTTF/MTTR	Exp(3000 min) / Exp(90 min)

Processing times and tool counts reflect a simplified teaching fab. Real fab recipes involve hundreds of steps; this problem focuses on the re-entrant bottleneck structure. Tool MTTF/MTTR should be sourced from equipment maintenance records.

Key Performance Indicators.

- Lot cycle time (release to test-complete), mean and 90th percentile
- Lithography tool utilization
- WIP level at lithography queue
- Hot-lot vs. normal-lot cycle-time ratio
- Throughput (lots completed per day)

Deliverables.

1. Baseline model; validate throughput and cycle time against Little's Law: $CT = WIP/TH$.
2. Warm-up analysis: determine how many lot completions are needed before steady-state metrics stabilize.
3. Bottleneck relief: increase lithography from 2 to 3 tools; quantify cycle-time reduction and new bottleneck location.
4. Hot-lot impact: vary hot-lot fraction from 0% to 20%; report normal-lot cycle-time penalty per percentage point of hot lots.
5. WIP control: implement CONWIP (release a new lot only when one exits test); compare steady-state cycle time and throughput to unconstrained release.

Extensions.

1. Implement preventive maintenance for lithography (4-hour PM every 5 days); evaluate whether an off-peak PM window reduces cycle-time variance.
2. Model a reticle defect: 5% of lots at the second lithography visit require rework (return to step 2); evaluate cycle-time and throughput impact.

Research Literature. Search WSC, *IEEE Transactions on Semiconductor Manufacturing*, and *IIE Transactions* for “re-entrant queue simulation”, “CONWIP semiconductor DES”, and “bottleneck analysis wafer fab”. Identify the theoretical results on instability in re-entrant queues (Kumar 1993; Lu, Kumar, and Seidman 1994) and understand what the decomposition approximation assumes about independence across visits.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: WIP per step, lithography queue depth (visits 1, 2, 3 separately), hot-lot count, CONWIP token count, cycle-time moving average.

Action $\mathcal{A}$: CONWIP cap $\{4, 6, 8, 10, 12\}$ and dispatching priority at lithography: $\{\text{FIFO}, \text{SRPT}, \text{hot-first}\}$.

Decision epochs: at each lot completion.

Reward r_t : $\text{throughput_rate} - \alpha \cdot \text{cycle_time} - \beta \cdot \text{hot_lot_sla_violations}$.

RL deliverables.

1. Train PPO or SAC; compare to CONWIP + FIFO baseline on throughput and mean cycle time.
2. Study interpretability: at which WIP level does the agent switch from FIFO to hot-lot priority? Compare this threshold to the analytical prediction from the bottleneck model.

Research Track Deliverables (Track B).

1. **Research hypothesis.** E.g., “The decomposition-based cycle-time approximation underestimates mean cycle time by more than 15 % when lithography utilization exceeds 0.85.”
2. **Analytical benchmark.** Derive the decomposition approximation (treat each visit as an independent M/G/1 queue); compute mean cycle time across utilization levels $\rho_{\text{litho}} \in \{0.70, 0.80, 0.90, 0.95\}$.
3. **Factorial comparison.** Simulate at the same utilization levels (adjust inter-arrival rate); report the simulation vs. approximation discrepancy and its sources (visit correlation, finite buffers).
4. **CONWIP vs. unlimited WIP.** Run a full experiment across CONWIP caps $\{4, 6, 8, 10, \infty\}$; fit a response surface for cycle time vs. cap; identify the Pareto front of (throughput, cycle time).
5. **Generalizable conclusions.** Express findings as: “For re-entrant systems with k visits to a bottleneck at utilization ρ , the decomposition error is approximately $f(\rho, k)$.”

EP07 — Perishable Food Processing Line

Domain: Manufacturing • **Difficulty:** ★★ • **Key techniques:** Batch processing, perishability constraints, throughput bottleneck analysis

System Overview. A fresh-produce packing facility operates a single 16-hour shift processing bell peppers received from field trucks. Product must reach cold storage within three hours of unloading or it is wasted. The facility must satisfy random retail orders while minimizing both waste and order shortfalls.

Operational Description. Trucks arrive stochastically and unload peppers into a staging area. Product flows through four sequential stages: batch washing, conveyor inspection, manual sorting (three stations), and automated packing (two lines). Cosmetic rejects are removed at inspection. Packed product moves immediately to cold storage. Customer orders arrive throughout the day and are fulfilled from cold storage if inventory is available; otherwise the order is short.

Data.

Parameter	Value
Truck inter-arrival time	Exp(45 min)
Load per truck	800 kg $\pm$ 10% (uniform)
Washing	batch 200 kg, 15 min/batch
Inspection	conveyor; 0.5 kg/min; rejects 8%
Sorting stations	3; $\mathcal{N}$ (4 min, 0.5 min) per 16-kg box
Packing lines	2 automated; 40 boxes/hr each
Cold-storage capacity	2,000 kg
Perishability window	3 hr from unload to cold storage
Order inter-arrival	Exp(160 min) (6 orders/day on average)
Order size	500 kg
Order pickup window	$\pm$ 4 hr from order arrival
Shortage penalty	\$50/order not filled

Truck arrival rates and load variability should be measured at the receiving dock. Rejection rates vary by harvest season and field conditions.

Key Performance Indicators.

- Shift throughput (kg processed and packed)
- Waste rate (% by weight exceeding the 3-hour window)
- Order fill rate (% orders fully satisfied)
- Stage-level utilization and queue lengths
- Cold-storage peak occupancy

Deliverables.

1. Simulate 30 shifts; identify the primary throughput bottleneck.
2. Batch-size sensitivity at washing: test 100, 200, and 300 kg; report throughput and waste rate for each.
3. Evaluate adding a 4th sorting station; quantify marginal improvement.

4. Demand surge: double order arrivals for 5 consecutive days; report shortfall count and cold-storage overflow events.
5. Model partial spoilage: for product exceeding a 2.5-hour processing delay, apply exponential quality decay and compute economic waste cost.

Extensions.

1. Replace stochastic truck arrivals with an appointment schedule; optimize slot spacing to minimize both idle washing capacity and perishability risk.
2. Model a cold-storage breakdown lasting 2 hours (1% probability per shift); develop a contingency plan using refrigerated staging.

Research Literature. Search WSC, *European Journal of Operational Research*, and *Food Control* for “perishable food production simulation”, “food processing line DES”, and “quality-throughput tradeoff perishables”. Identify whether speed–yield tradeoffs in food processing lines have been modeled jointly with quality-inspection policies in the DES literature.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: line speed (current level), buffer fill levels between stages, mean product age in each buffer, recent defect rate (rolling 15-min window).

Action $\mathcal{A}$: line speed {low, med, high} and quality inspection trigger {on, off} (6 joint actions).

Decision epochs: every 30 minutes.

Reward r_t : $\text{yield_kg_per_hr} - \alpha \cdot \text{waste_cost} - \beta \cdot \text{quality_violation_penalty}$.

RL deliverables.

1. Train DQN; compare to the fixed-speed, always-inspect baseline on yield and waste.
2. Interpret policy: does the agent reduce speed proactively when buffer age exceeds a threshold, or reactively after defect rate rises?

EP08 — Pharmaceutical Batch Manufacturing with Changeovers

Domain: Manufacturing • **Difficulty:** *** • **Key techniques:** Batch scheduling, sequence-dependent setups, OEE, supply reliability

System Overview. A GMP-certified pharmaceutical plant produces four drug products in three shared reactors. Products must be manufactured in campaign blocks; switching between products requires cleaning and changeover. Quality control (QC) testing follows every batch, and failures trigger costly rework. The goal is to maximize equipment effectiveness while meeting weekly demand.

Operational Description. Batches are assigned to reactors according to a campaign schedule. When a reactor completes a batch, it either begins a same-product batch (minimal cleaning) or undergoes a sequence-dependent changeover before the next product. Each completed batch enters a QC hold queue; failed batches return to the reactor for rework. Reactors are subject to random breakdowns.

Data.

Changeover matrix (hours; symmetric):

From \ To	A	B	C	D
A	1	4	6	8
B	4	1	5	7
C	6	5	1	3
D	8	7	3	1

Parameter	Value
Reactors	3
Batch processing time	$\mathcal{N}(18, 2)$ hr (A), $\mathcal{N}(24, 3)$ hr (B), $\mathcal{N}(12, 1.5)$ hr (C), $\mathcal{N}(30, 4)$ hr (D)
Weekly demand (batches)	A: 3, B: 2, C: 4, D: 1 (each ± 1 stochastic)
QC testing time	$\text{Exp}(8 \text{ hr})$
QC failure rate	5%
Rework	full reprocess (same batch time, no additional changeover)
Reactor MTTF	$\text{Exp}(500 \text{ hr})$
Reactor MTTR	$\text{Exp}(6 \text{ hr})$

Changeover times come from validated cleaning procedures and should be confirmed with manufacturing science. QC failure rates vary by product and lot history; 5% is a conservative estimate for a mature product.

Key Performance Indicators.

- On-time delivery rate (batches delivered within the weekly horizon)

- Overall Equipment Effectiveness (OEE = availability $\times$ performance $\times$ quality rate) per reactor
- Mean changeover time per product campaign
- Product inventory level at each staging area
- Rework rate

Deliverables.

1. Simulate 52 weeks; compute OEE per reactor under a fixed $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ rotation.
2. Compare two campaign sequencing heuristics: (a) fixed rotation as above; (b) nearest-neighbor (minimize next changeover time); report OEE and on-time delivery.
3. QC sensitivity: vary failure rate from 2% to 10%; report delivery-rate degradation.
4. Evaluate a 4th reactor: quantify throughput gain, OEE change, and break-even against capital cost.
5. Report 95% CIs for OEE and on-time delivery across 30 replications.

Extensions.

1. Add a QC lab capacity constraint (only 2 batches in QC simultaneously); evaluate effect on reactor idle time.
2. Simulate a raw-material supply disruption for product C lasting 14 days; assess schedule resequencing options and their delivery impact.

Research Literature. Search WSC, *Computers & Chemical Engineering*, and *Journal of Pharmaceutical Innovation* for “pharmaceutical batch scheduling simulation”, “changeover optimization DES”, and “clean-in-place campaign scheduling”. Understand sequence-dependent changeover minimization and whether prior work has jointly optimized batch sequencing with regulatory CIP compliance.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: equipment availability per unit, batch queue (product type and size), time since last CIP per equipment, time to next regulatory deadline.

Action $\mathcal{A}$: select next batch for each available equipment unit; decide CIP now vs. defer.

Decision epochs: at each batch completion.

Reward r_t : $-\text{total_changeover_time} - \alpha \cdot \text{schedule_delay} - \beta \cdot \text{regulatory_violation_count}$.

RL deliverables.

1. Train PPO; compare to the shortest-changeover heuristic (greedy CIP deferral) on total campaign duration and CIP frequency.
2. Evaluate policy robustness under demand surge (3 urgent batches injected mid-campaign); report recovery time.

3 Retail and Service Operations

EP09 — Coffee Shop Drive-Through and Walk-In Operations

Domain: Retail / Service • **Difficulty:** ★★ • **Key techniques:** Dual-channel queueing, non-stationary demand, balking, staffing

System Overview. A drive-through coffee shop serves customers through two channels: a single-lane drive-through window and a walk-in counter staffed by two baristas. Demand peaks sharply in the morning and at lunch. Customers balk when queues are perceived as too long, leading to lost revenue.

Operational Description. Drive-through and walk-in arrival rates are split proportionally by time period. Drive-through customers observe the queue length before entering the lane and may balk. Walk-in customers estimate visible wait time before joining. A fraction of orders are “complex” (e.g., custom espresso drinks) and take additional preparation time. Mobile-order customers are a candidate intervention that reduces service time for that subgroup.

Data.

Period	Hours	Total arrival rate	Channel split
Morning rush	06–09	90/hr	60% DT, 40% WI
Midmorning	09–12	40/hr	50% DT, 50% WI
Lunch	12–14	65/hr	55% DT, 45% WI
Afternoon	14–17	30/hr	45% DT, 55% WI
Evening	17–20	50/hr	55% DT, 45% WI

Parameter	Value
Drive-through service time	LN(3.5 min, 1.2 min)
Walk-in service time	LN(2.8 min, 1.0 min)
Complex order fraction	20%; adds 2 min to service
DT lane capacity	10 cars
DT balking	25% balk if ≥ 8 cars; 100% if lane full
WI balking	30% balk if estimated wait ≥ 12 min
Revenue per customer	\$6

Arrival rates and service times should be measured by observational time-study during representative shifts. Balking probabilities can be estimated from queue-length vs. join-rate data.

Key Performance Indicators.

- Mean service time and queue length by channel and period
- Balking/lost customer rate by channel
- Channel utilization
- Daily lost revenue from balking

Deliverables.

1. Simulate 30 operating days; report peak-hour queue lengths and balking rates.
2. Add a mobile barista during the two peak periods; evaluate queue-length reduction and utilization.
3. Mobile ordering channel: 30% of walk-in customers place mobile orders (service time reduced to 1.5 min for that sub-group); evaluate wait-time improvement.
4. Drive-through capacity: extend lane to 15 cars; evaluate balking-rate change and lost-revenue recovery.
5. Revenue analysis: estimate daily lost revenue under baseline; compare to best combined intervention using CRN-paired confidence intervals.

Extensions.

1. Model staff breaks (each barista: two 15-minute breaks per shift at random times); evaluate service-continuity impact.
2. Add day-of-week demand variation: Mondays and Fridays are 20% above average; calibrate staffing accordingly.

Research Literature. Search WSC, *International Journal of Hospitality Management*, and *Service Science* for “drive-through queue simulation”, “service operations balking DES”, and “mobile ordering impact”. Identify how balking probability curves are estimated from empirical queue-length data and whether multi-channel coffee-shop models have been studied in the simulation literature.

Advanced RL Extension (Track C). MDP formulation.

State S : drive-through queue length, walk-in queue length, current demand period, active barista count.

Action A : deploy extra barista $\{0, 1\}$ and mobile-order promotion $\{0, 1\}$ (4 joint actions).

Decision epochs: every 30 minutes (period start).

Reward r_t : $\text{revenue_from_served} - \alpha \cdot \text{idle_staff_cost} - \beta \cdot \text{balk_count} - \text{revenue_per_customer}$.

RL deliverables.

1. Train DQN; compare to peak-only staffing heuristic on daily revenue and staff cost.
2. Evaluate agent on held-out holiday week with unusual demand; report out-of-distribution performance degradation.

EP10 — Supermarket Checkout Lane Design

Domain: Retail / Service • **Difficulty:** ★★ • **Key techniques:** Multi-server queues, express lanes, self-checkout, lane choice modeling

System Overview. A suburban supermarket operates 08:00–22:00 with 14 checkout lanes: 8 full-service, 4 self-checkout kiosks, and 2 express lanes (≤ 15 items). The study evaluates staffing levels, express-lane item limits, and self-checkout expansion to minimize customer wait times during peak hours.

Operational Description. Customers select a lane based on visible queue lengths and eligibility (item count for express; preference for self-checkout). Self-checkout transactions incur an error rate requiring attendant intervention. Two attendants each supervise two kiosks. Cashier breaks dynamically reduce capacity.

Data.

Parameter	Value
Arrival rate	60/hr (08–11), 90/hr (11–17), 180/hr (17–19), 90/hr (19–22)
Full-service item count	PERT(10, 45, 120)
Express item count	Uniform(1, 15)
Self-checkout item count	Uniform(3, 25)
Full-service scan rate	24 items/min; add 20% for bagging
Express scan rate	28 items/min
Self-checkout scan rate	14 items/min including assisted flow
Self-checkout error rate	12%; attendant adds Exp(2 min)
Lane choice (self-checkout)	40% of eligible customers prefer self-checkout
Staff	8 full-service cashiers; 2 express cashiers; 2 SCO attendants

*Arrival rates and item distributions should be drawn from point-of-sale transaction records.
Self-checkout error rates vary with customer age and product mix.*

Key Performance Indicators.

- Mean wait time by lane type
- Queue length at peak (95th percentile)
- Lane utilization
- Self-checkout error penalty time per transaction
- Customer throughput (customers/hour)

Deliverables.

1. Simulate 60 days; verify Little's Law for each lane type.
2. Peak staffing: open 2 additional full-service lanes during 17–19; evaluate wait-time reduction and added labor cost (\$18/hr/cashier).
3. Self-checkout expansion: replace 2 full-service lanes with 4 additional kiosks; report total average wait time (all customers combined).
4. Express limit: change from 15 to 20 items; evaluate spillover effect on full-service queues.
5. Report 95% CIs for peak wait time under baseline and best alternative (30 replications).

Extensions.

1. Model cashier breaks (15 min every 2 hours); dynamically close the lane and redirect customers; evaluate wait-time spike.
2. Introduce scan-and-go for 10% of customers (bypass checkout); add a spot-check process (1% of scan-and-go customers; $\text{Exp}(5 \text{ min})$); evaluate net throughput impact.

Research Literature. Search WSC and *Management Science* for “supermarket checkout simulation”, “self-checkout adoption DES”, and “lane-choice behavior queueing model”. Understand the Multinomial-Logit lane-choice models used in the queueing literature and whether self-checkout throughput variability has been measured empirically.

Advanced RL Extension (Track C). MDP formulation.

State S : queue length per lane type, average basket size in current wave, time of day, staff-on-shift count.

Action A : number of each lane type open: express $\{0, \dots, 4\}$, regular $\{0, \dots, 8\}$, self-checkout $\{0, \dots, 4\}$.

Decision epochs: every 15 minutes.

Reward r_t : $-\text{mean_customer_wait} - \alpha \cdot \text{idle_cashier_cost}$.

RL deliverables.

1. Train PPO; compare to the manager heuristic (open a lane when queue > 5) on mean wait and total labor cost.
2. Analyze how the agent trades off self-checkout vs. cashier lanes as basket-size distribution shifts.

EP11 — E-Commerce Order Fulfillment Center

Domain: Retail / Service • **Difficulty:** *** • **Key techniques:** Multi-stage flow, SLA compliance, batch picking, peak-season scaling

System Overview. A single-site e-commerce fulfillment center processes 15,000 orders per day during normal operations and 30,000 orders per day during a 14-day peak season. Orders must be shipped within 4 hours of receipt (SLA). The study identifies which stage first bottlenecks under peak load and evaluates operational interventions.

Operational Description. Orders arrive continuously. Each order is assigned to 1–3 pick zones; pickers travel through their zone, collect items, and route totes to packing. Packers produce one or more cartons per order. Cartons travel on a conveyor sorter to assigned shipping docks where trailers depart on a fixed 90-minute cycle. Orders not loaded before trailer departure miss the SLA.

Data.

Parameter	Value
Order arrival rate (normal)	15,000/day (Exp(5.76 sec) inter-arrival)
Order arrival rate (peak)	30,000/day
Pick zones	4; order visits 1–3 zones: probs 40%, 40%, 20%
Picker travel time per zone visit	LN(8 min, 3 min)
Packing time (single carton)	$\mathcal{N}$ (2 min, 0.5 min)
Packing time (multi-carton, 30%)	$\mathcal{N}$ (5 min, 1.5 min)
Sorter throughput	4,000 units/hr
Sorter jam frequency	Pois(0.5)/hr; repair Exp(10 min)
Trailer departure schedule	every 90 min
Docks	4
Pickers/zone/shift	20
Packers	10
SLA threshold	4 hr from order receipt to trailer departure

Picker travel times depend heavily on zone layout, slotting strategy, and order-profile item density. These should be measured via time-and-motion study and warehouse management system (WMS) logs.

Key Performance Indicators.

- SLA compliance rate (target $\geq 95\%$)
- Order cycle time (receipt to dock departure)
- Pick-zone, packer, and sorter utilization
- Trailer departure fullness (TEU equivalent)
- Sorter jam impact on downstream throughput

Deliverables.

1. Simulate 30 normal days; validate SLA compliance.
2. Peak season (14 days): identify which stage first breaches SLA at $2\times$ demand.

3. Add 25% more pickers during peak; report SLA recovery and incremental labor cost.
4. Batch picking: consolidate picks from 3 orders in one zone trip, reducing travel time by 40%; evaluate SLA improvement and packer queue depth.
5. Cross-dock option: for multi-carton orders with vendor pre-packing (20% of orders), bypass the pack stage; evaluate SLA and sorter throughput.

Extensions.

1. Add autonomous mobile robots (AMR) to Zone 1: pick time reduced to $\mathcal{N}(3, 0.5)$ min; evaluate SLA improvement and payback period (\$250K/robot, 3 robots, 5-year horizon).
2. Model returns processing: 8% of shipped orders generate a return within 7 days; returns enter the pack stage for inspection at 5 min each; evaluate impact on forward-flow SLA.

Research Literature. Search WSC, *International Journal of Production Economics*, and *European Journal of Operational Research* for “order fulfillment simulation”, “warehouse picker routing DES”, and “batch picking optimization”. Identify whether wave-release timing and batch-size co-optimization have been studied in simulation settings.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: order queue depth by pick zone, pickers available, packing-station utilization, time remaining in shift.

Action $\mathcal{A}$: batch release size $\{5, 10, 20, 30\}$ and routing strategy $\{\text{zone, combined, wave}\}$.

Decision epochs: at each picker-becomes-idle event.

Reward r_t : $\text{orders_shipped_on_time} - \alpha \cdot \text{picker_idle_fraction} - \beta \cdot \text{overtime_hours}$.

RL deliverables.

1. Train DQN; compare to fixed-batch zone-picking on on-time rate and labor cost.
 2. Evaluate under holiday demand surge (+80%); report whether the learned policy degrades gracefully.
-

EP12 — Bank Branch with Teller and Digital Alternatives

Domain: Retail / Service • **Difficulty:** ★★ • **Key techniques:** Priority queues, multi-server, reneging, ATM failure, digital adoption

System Overview. A suburban bank branch serves walk-in customers through three channels: teller windows (cash transactions), customer service representative (CSR) desks (account changes and loans), and ATMs (cash and balance inquiries). ATM downtime deflects customers to tellers, increasing wait times. Reneging customers represent lost revenue.

Operational Description. Customers arrive and self-select a channel based on transaction type. Teller customers join a single shared queue; CSR customers choose walk-in or appointment. ATM customers join a short queue at the machine; when the ATM is out of service, ATM-type customers switch to the teller queue. Customers renege from teller and CSR queues if their wait exceeds a tolerance threshold.

Parameter	Value
Daily arrivals (weekday)	80; weekend (Sat) 120
Arrival peak	11:00–13:00 (2× hourly average)
Channel split	60% teller, 25% CSR, 15% ATM
Teller service time	Exp(5 min)
CSR walk-in service time	$\mathcal{N}$ (20 min, 5 min)
CSR appointment service time	$\mathcal{N}$ (30 min, 10 min)
ATM service time	Exp(2 min)
ATM out-of-service probability	5%/day; repair Exp(3 hr)
Tellers	3 (weekday), 2 (Saturday)
CSRs	2 (weekday), 1 (Saturday)
Teller reneging threshold	Exp(20 min)
CSR reneging threshold	Exp(30 min)
Average transaction value lost	\$45 per reneging customer

Arrival counts and service times should be sourced from branch management system logs. Reneging thresholds can be estimated from queue-length observation and exit surveys.

Key Performance Indicators.

- Mean wait time by channel
- Reneging rate and daily lost revenue
- Teller and CSR utilization
- ATM downtime impact (teller queue overflow events)
- Revenue recovered under each intervention

Deliverables.

1. Simulate 1 year; report utilization and wait time by channel.
2. ATM expansion: add a second ATM; evaluate teller-queue deflection and wait-time reduction.
3. CSR appointment scheduling: implement 50% appointment / 50% walk-in split; evaluate CSR idle time versus wait reduction.
4. Reneging sensitivity: vary the teller patience threshold from 10 to 30 min; identify the inflection point for daily lost revenue.
5. Saturday staffing: evaluate adding one teller; report 95% CI for Saturday wait-time improvement.

Extensions.

1. Digital-banking campaign: 20% of teller transactions shift to the ATM or mobile app; re-evaluate staffing adequacy.
2. Add a drive-through teller (2 bays); model channel choice between drive-through and walk-in; evaluate overall throughput.

Research Literature. Search WSC, *Management Science*, and *Operations Research* for “bank branch simulation”, “multi-channel service teller queueing”, and “digital channel adoption service operations”. Understand how channel substitution rates are modeled and what is known about teller-count sensitivity in Erlang-A (with abandonment) models.

Advanced RL Extension (Track C). MDP formulation.

State S : teller queue length, in-progress transaction count, time of day, digital-adoption rate (rolling 30-day average).

Action $\mathcal{A}$: number of tellers open $\{2, 3, 4, 5, 6\}$ and digital-promotion intensity $\{\text{low, med, high}\}$.

Decision epochs: every 30 minutes.

Reward r_t : $-\text{customer_wait} - \alpha \cdot \text{teller_idle_cost} + \beta \cdot \Delta \text{digital_adoption}$.

RL deliverables.

1. Train PPO; compare to fixed-schedule staffing on mean wait and labor cost.

2. Study the long-run adoption trajectory under the RL promotion policy vs. no promotion; quantify teller-count savings at steady-state adoption.

4 Logistics and Supply Chain

EP13 — Multi-Echelon Inventory Network

Domain: Logistics / Supply Chain • **Difficulty:** ★★★ • **Key techniques:** (s, S) policy, multi-echelon inventory, backorders, supply uncertainty

System Overview. A consumer-goods supply chain connects one upstream supplier, one distribution center (DC), and five retail stores. Each node operates an (s, S) replenishment policy. The study jointly optimizes DC and store policies to minimize total system cost (holding, ordering, and backorder penalties) while maintaining target fill rates.

Operational Description. Each store faces independent daily Poisson demand. When store inventory falls to or below s_i , an order of size $S_i - s_i$ is placed to the DC; the order arrives after a stochastic lead time. The DC places orders to the supplier when its position hits s_{DC} ; supplier lead time is also stochastic. The supplier fills DC orders in full if inventory is available; partial fills propagate downstream shortages.

Data.

Parameter	Value
Store demand	Pois(50) units/day per store (stores vary by $\pm 20\%$)
DC $\rightarrow$ store lead time	$\mathcal{N}$ (2 days, 0.5 days)
Supplier $\rightarrow$ DC lead time	$\mathcal{N}$ (7 days, 1.5 days)
Holding cost	\$0.50/unit/day (DC), \$1.00/unit/day (store)
Backorder penalty	\$10/unit/day (stores)
Ordering fixed cost	\$200 (DC order), \$50 (store order)
Supplier capacity	unlimited; partial fill if DC inventory short
Initial (s, S) at DC	(500, 1000)
Initial (s, S) per store	(100, 200)

Demand parameters should be estimated from point-of-sale data by SKU. Lead-time distributions should be extracted from purchase-order records. Holding and backorder costs require finance-team input on capital cost and service-agreement penalties.

Key Performance Indicators.

- Total system cost per day (holding + ordering + backorder)
- Fill rate at each store (target $\geq 97\%$)
- DC and store inventory levels (mean, 95th percentile)
- Backorder frequency and duration
- Order frequency per node

Deliverables.

1. Simulate 365 days with the initial policy; validate fill rate and cost breakdown.
2. Single-store optimization: fix the DC policy; perform a grid search over (s, S) for one store to minimize its cost subject to $\geq 97\%$ fill rate.
3. System-wide optimization: jointly optimize DC and all five store policies; report cost reduction versus the initial policy.
4. Lead-time sensitivity: double the DC lead-time variance; report fill-rate degradation.
5. Holiday surge: double demand for 14 days; report stockout duration and recovery time.

Extensions.

1. Lateral transshipment: allow stores to ship to each other at \$3/unit with a 1-day delay; evaluate as a fill-rate backstop.
2. Model a supplier shortage (capacity halved for 30 days); evaluate the safety-stock level at the DC needed to maintain a 99% fill rate.

Research Literature. Search *Operations Research*, *Management Science*, and WSC for “multi-echelon inventory simulation”, “ (s, S) policy optimization DES”, and “base-stock policy stochastic lead times”. The Clark-Scarf decomposition (1960) is the analytical benchmark; understand its assumptions and identify where stochastic lead times and partial fills invalidate it.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: on-hand inventory and pipeline orders at each node (DC + 5 stores), days since last order per node, recent backorder count.

Action $\mathcal{A}$: reorder parameters (s_i, S_i) per node, discretized into a finite set of candidate policies.

Decision epochs: daily (review each node's inventory position).

Reward r_t : $-\text{total_system_cost}$ (holding + ordering + backorder penalty).

RL deliverables.

1. Train PPO; compare to the simulation-optimized (s, S) policy and to the Clark-Scarf decomposition solution on total cost and fill rates.
2. Evaluate robustness: test the trained policy when supplier lead-time variance doubles; does the RL policy self-adapt better than a fixed (s, S) ?
3. Interpret the learned policy: does the agent behave as a base-stock policy at the DC? Use regression on logged state-action pairs to characterize the implicit policy structure.

Research Track Deliverables (Track B).

1. **Research hypothesis.** E.g., “Under lead-time $CV > 0.5$, the (s, S) policy requires $> 20\%$ more safety stock than a base-stock policy to achieve a 97% fill rate.”
2. **Analytical benchmark.** Compute the Clark-Scarf base-stock solution assuming independent stages; report the fill rate and cost it predicts vs. the simulation result for the same policy.
3. **Factorial experiment.** 3×3 design: lead-time $CV \{0.2, 0.5, 0.8\} \times$ backorder penalty $\{5, 10, 20\}$ \$/unit; report optimal (s, S) and total cost in each cell.
4. **Policy comparison.** Compare (s, S) to a pure base-stock (s, ∞) policy under each condition; quantify when the fixed-order-quantity discipline of (s, S) dominates.
5. **Generalizable conclusions.** Identify the lead-time uncertainty threshold above which simulation-based optimization outperforms Clark-Scarf by more than 5% in total cost.

EP14 — Container Port Terminal Operations

Domain: Logistics / Transportation • **Difficulty:** *** • **Key techniques:** Ship arrivals, berth allocation, resource contention, demurrage costs

System Overview. A medium-sized container terminal handles bulk container traffic through three berths, four quay cranes, twelve yard trucks, and three yard cranes. Ships waiting for a berth incur demurrage costs. The study evaluates quay crane and truck counts, berth scheduling rules, and yard crane reliability.

Operational Description. Ships arrive stochastically and queue at anchor if all berths are occupied. Upon berthing, two quay cranes work in parallel to unload and load containers. Yard trucks transfer containers between quay and yard. Yard cranes stack and retrieve containers. Ship departure is triggered when all container moves are complete.

Data.

Parameter	Value
Ship inter-arrival time	Exp(18 hr)
Ship size	Uniform(800, 2400) TEU
Berths	3; one ship per berth
Quay cranes	4 (2 assigned per berth while in use)
Quay crane cycle time	Exp(2.5 min)/container move
Yard trucks	12; travel time Exp(15 min) round-trip
Yard cranes	3; stack time Exp(3 min)/container
Yard utilization (initial)	60% of 8,000 TEU capacity
Demurrage cost	\$5,000/hr waiting at anchor

Ship inter-arrival times and size distributions should be sourced from the port’s vessel call records. Crane cycle times should be measured from terminal operating system (TOS) event logs.

Key Performance Indicators.

- Ship turnaround time (berth arrival to departure)
- Ship waiting time at anchor and daily demurrage cost
- Quay crane, truck, and yard crane utilization
- Yard occupancy (% of 8,000 TEU)
- Daily TEU throughput

Deliverables.

1. Simulate 365 days; validate turnaround time against the 24–36-hour benchmark for this terminal class.

2. Resource sensitivity: vary quay cranes (3–5), trucks (8–16), yard cranes (2–4); report turnaround time and utilization.
3. Berth scheduling: implement priority allocation for large ships ($> 2,000$ TEU); compare to FCFS on turnaround and demurrage.
4. Yard crane breakdown: simulate one crane offline for 8 hr (probability 1%/day); report throughput impact.
5. Report 95% CIs for annual demurrage cost under baseline and optimal configuration.

Extensions.

1. Add night-shift productivity reduction (20% lower crane rate on 21:00–06:00 shifts); evaluate turnaround-time increase and staffing cost of equalizing productivity.
2. Model a seasonal trade peak (summer volume 30% above average); evaluate berth expansion during peak season versus year-round.

Research Literature. Search WSC, *Transportation Science*, and *European Journal of Operational Research* for “container terminal simulation”, “berth allocation DES”, and “quay crane scheduling port operations”. Understand the benchmark 24–36-hour turnaround standard and whether prior simulation studies have compared priority-based vs. FCFS berth allocation rigorously.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: ships at anchor (count and size class), berths occupied, cranes assigned per berth, yard occupancy fraction.

Action $\mathcal{A}$: berth assignment rule for the arriving ship: {FCFS, large-first, small-first} and crane reallocation {2, 3} cranes per active berth.

Decision epochs: at each ship arrival and berth completion.

Reward r_t : $-\text{demurrage_cost_per_hr} + \alpha \cdot \text{TEU_throughput_per_day}$.

RL deliverables.

1. Train DQN; compare to FCFS on total annual demurrage and TEU throughput.
 2. Evaluate under congestion (+30% arrival rate); report whether RL advantage over FCFS grows at higher load.
-

EP15 — Last-Mile Parcel Delivery

Domain: Logistics / Transportation • **Difficulty:** ★★ • **Key techniques:** Vehicle routing, failed deliveries, fleet sizing, demand surge

System Overview. A single urban depot serves 30 delivery zones across a 30 km × 30 km grid using a fleet of 12 vehicles. Each vehicle must complete its route and return to the depot by 19:00. Failed deliveries (absent recipients) are re-attempted the next day, driving backlog and cost.

Operational Description. Each morning the depot loads 1,500 parcels and dispatches vehicles. Vehicles follow pre-planned routes; at each stop, the delivery outcome is stochastic (successful or failed). Failed parcels return to the depot and enter the next day’s manifest. Vehicles that cannot complete their routes by 19:00 leave undelivered stops as backlog.

Parameter	Value
Daily parcels	1,500 (normal); 2,250 (+50% surge, 7-day period)
Delivery zones	30; each zone Poisson(50) parcels/day
Fleet size	12 vehicles; capacity 150 parcels each
Shift	07:00–19:00 (12 hr)
Delivery time per stop	$\mathcal{N}(3 \text{ min}, 1 \text{ min})$
Inter-stop travel time	$\mathcal{N}(3 \text{ min}, 1 \text{ min})$ (zone-average)
Failed delivery rate	15% of residential stops
Re-delivery added cost	\$3.50/parcel
Fuel cost	\$0.25/km
Vehicle depreciation	\$15/vehicle-day
Average inter-stop distance	1.2 km

Failed delivery rates vary by delivery time, zone type (residential vs. commercial), and carrier notification practices. Vehicle cost parameters should be obtained from fleet management records.

Key Performance Indicators.

- Same-day delivery rate (% of parcels delivered on day of assignment)
- Failed delivery rate and re-delivery cost per parcel

- Vehicle utilization (parcels/vehicle/day)
- Daily cost per parcel (fuel + depreciation + re-delivery)
- Overtime events (vehicles returning after 19:00)

Deliverables.

1. Simulate 60 days; validate parcels delivered per day against fleet capacity.
2. Delivery notification system: reduces failure rate from 15% to 8%; report cost savings versus implementation cost (\$10K/month for SMS system).
3. Route optimization: reduce inter-stop travel by 20%; evaluate whether fleet can be reduced by 1 vehicle without increasing backlog.
4. Cargo bike deployment: 2 electric cargo bikes for the two densest zones (capacity 30 parcels, operating cost \$5/day); evaluate cost-per-parcel reduction in those zones.
5. Demand surge: simulate the 7-day +50% surge; report backlog buildup and days until backlog clears after surge ends.

Extensions.

1. Vehicle breakdown: 2% daily breakdown probability; 4-hour repair wait or replacement from a shared pool; evaluate backlog impact.
2. Time-window constraints: 30% of parcels require delivery between 09:00–12:00; restructure routes and report impact on vehicle count.

Research Literature. Search WSC, *Transportation Research Part E*, and *European Journal of Operational Research* for “last-mile delivery simulation”, “vehicle routing failed deliveries DES”, and “urban parcel delivery fleet sizing”. Identify what is known about the interaction of failed-delivery re-attempt cycles and same-day delivery rate.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: parcel backlog per zone, vehicles available, time of day, failed-delivery backlog count.

Action $\mathcal{A}$: zone priority ranking and whether to deploy an extra vehicle from depot reserve $\{0, 1\}$.

Decision epochs: each morning dispatch and after major failure event.

Reward r_t : same-day_delivery_rate $- \alpha \cdot$ re-delivery_cost $- \beta \cdot$ overtime_hours.

RL deliverables.

1. Train PPO; compare to equal-priority routing on same-day rate and cost.
2. Analyze learned zone prioritization: does the agent favor low-failure-rate zones or high-density zones? Discuss implications for real dispatch practice.

EP16 — Cold-Chain Pharmaceutical Distribution

Domain: Logistics / Healthcare • **Difficulty:** *** • **Key techniques:** Temperature-sensitive inventory, compliance risk, multi-customer replenishment

System Overview. A pharmaceutical distributor maintains a temperature-controlled warehouse serving 8 hospital pharmacies (daily delivery) and 30 retail pharmacies (twice-weekly delivery). Three temperature-sensitive products (vaccines, biologics, and oncology drugs) must be stored at 2–8°C. Temperature excursions during transport cause product quarantine, testing, and potential disposal.

Operational Description. The warehouse replenishes from the manufacturer and dispatches orders to customers via two refrigerated trucks. Each trip has a small probability of a temperature excursion; excursed product enters a quarantine-and-test process and may be destroyed. The inventory policy must balance holding cost, backorder penalty, and disposal cost.

Data.

Parameter	Value
Products	3 (vaccines, biologics, oncology drugs)
Hospital demand	Pois(20) units/pharmacy/day per product
Retail demand	Pois(8) units/pharmacy/day per product
Manufacturer lead time	Exp(3 days); min. order 100 units
Lot size	500 units
Refrigerated trucks	2
Hospital delivery frequency	daily
Retail delivery frequency	twice/week
Temperature excursion probability	2%/trip
Excursion test result (fail & destroy)	30% of excursed lots
Holding cost	\$2/unit/day
Backorder penalty	\$50/unit (hospital), \$20/unit (retail)
Disposal cost	\$150/unit

Excursion probabilities should be validated from temperature-logger records. Backorder penalties should reflect contractual service-level agreements with hospital clients.

Key Performance Indicators.

- Fill rate by customer type (target: $\geq 99\%$ hospital, $\geq 95\%$ retail)
- Temperature excursion and disposal rate
- Total logistics cost (holding + transport + backorder + disposal)
- Warehouse inventory level (mean and peak)
- Supplier stockout risk (days with zero warehouse inventory)

Deliverables.

1. Simulate 180 days; validate fill rates and waste rate.
2. Inventory optimization: find (s, S) per product and customer class that minimizes total cost subject to fill-rate constraints.
3. IoT temperature monitoring: detect excursions within 15 min (vs. end-of-trip); reduces excursion damage to partial-lot loss (50% instead of full lot); evaluate waste reduction and cost saving.
4. Delivery frequency increase (hospitals twice daily, retail three times/week): evaluate backorder reduction versus added transportation cost (\$200/additional trip).
5. Supply disruption: manufacturer halts shipment for 10 days; compute the safety-stock level at the warehouse needed to maintain hospital fill rate.

Extensions.

1. Seasonal demand spike (flu season: vaccine demand +40% for 60 days); evaluate pre-season inventory build-up strategy.
2. Add a second warehouse in a geographically separate city; split demand by region; evaluate total system cost and resilience.

Research Literature. Search WSC, *Supply Chain Management: An International Journal*, and *Journal of Pharmaceutical Sciences* for “cold chain simulation”, “temperature-sensitive inventory DES”, and “pharmaceutical distribution compliance risk”. Identify whether prior work has jointly modeled excursion risk, inventory policy, and delivery frequency.

Advanced RL Extension (Track C). MDP formulation.

State S : on-hand inventory per product and customer class, pipeline orders, cumulative excursion count this month, time to next scheduled delivery.

Action $\mathcal{A}$: order quantity per product (multiples of minimum order size) and delivery frequency adjustment {standard, extra-hospital-trip}.

Decision epochs: daily.

Reward r_t : $-\text{total_logistics_cost}$ (holding + transport + backorder + disposal).

RL deliverables.

1. Train DQN; compare to the simulation-optimized (s, S) policy on total cost and fill rate.
2. Evaluate under a supply disruption (manufacturer halt for 10 days); report whether the RL agent pre-builds inventory better than (s, S) .

5 Transportation Systems

EP17 — Highway Toll Plaza Congestion

Domain: Transportation • **Difficulty:** ★★ • **Key techniques:** Multi-server queueing, lane-choice behavior, all-electronic tolling

System Overview. An 8-lane toll plaza on a major interstate handles a mix of cash-paying, electronic-toll (E-ZPass), and commercial-truck traffic. The study evaluates lane conversion from cash to electronic, all-electronic tolling (AET) adoption, and congestion pricing as mechanisms to reduce vehicle delay.

Operational Description. Vehicles arrive as a Poisson process; each selects a lane based on observable queue length and eligibility. Trucks are restricted to the designated truck lane. E-ZPass vehicles prefer electronic lanes but switch if queue exceeds a threshold. The supervisor lane handles toll disputes, E-ZPass read failures, and other violations requiring manual processing; affected vehicles are redirected there from their original lane. Separately, *supervisor events*—equipment jams, detection errors, or lane blockages at cash or E-ZPass lanes—temporarily block the affected lane until the supervisor resolves the incident (the supervisor is modeled as a single shared resource with capacity 1; events at different lanes queue for the supervisor’s attention).

Data.

Parameter	Value
Lanes	3 cash, 3 E-ZPass, 1 truck, 1 supervisor
Arrival rate	1,200 veh/hr (peak 07–09, 17–19); 400 veh/hr (off-peak)
Vehicle mix	55% E-ZPass, 35% cash, 10% truck
E-ZPass service time	Exp(4 sec)
Cash service time	$\mathcal{N}$ (28 sec, 8 sec)
Truck service time	$\mathcal{N}$ (45 sec, 12 sec)
E-ZPass lane-switching threshold	switch if queue ≥ 4 cars
Supervisor lane vehicles	2% of all vehicles redirected for manual processing; Exp(5 min) service time
Supervisor events (lane blockage)	Pois(5)/hr at a randomly chosen active cash or E-ZPass lane; blocked for Exp(8 min) per event
Tolls	Cash \$3.00; E-ZPass \$2.50; Truck \$7.50

Traffic volumes and vehicle-mix data should be sourced from the toll authority's traffic counting system. Service-time distributions can be measured from surveillance video. The 2% supervisor-redirect fraction and supervisor event rate should be estimated from toll-transaction exception logs; these are the parameters most sensitive to plaza-specific enforcement practice.

Key Performance Indicators.

- Mean delay per vehicle by type
- Queue length at peak (vehicles, 95th percentile)
- Lane utilization
- Revenue per hour
- Violation rate and estimated fine recovery

Deliverables.

1. Simulate 365 days with peak/off-peak cycles; validate throughput.
2. Lane conversion: replace 1 cash lane with E-ZPass; report cash-vehicle delay change.
3. AET scenario: eliminate all cash booths; model a 6-month adoption ramp from 35% to 90% E-ZPass; report delay and revenue during transition.
4. Congestion pricing: raise E-ZPass toll to \$4.00 at peak, causing 10% of vehicles to shift to cash or off-peak travel; evaluate queue and daily revenue.

- Report 95% CIs for daily revenue and average vehicle delay under baseline and AET scenarios.

Extensions.

- Incident scenario: one cash lane fails for 2 hours (equipment breakdown); model queue spillback onto the highway approach.
- Truck weigh-in: 3% of trucks trigger a weigh-in station (+10 min delay); evaluate impact on truck-lane queue and commercial-vehicle delay costs.

Research Literature. Search WSC, *Transportation Research Part B*, and *Networks and Spatial Economics* for “toll plaza simulation”, “congestion pricing DES”, and “all-electronic tolling transition model”. Understand the equilibrium lane-choice models in the analytical literature and where DES adds value over fluid models for toll-plaza design.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: queue length per lane, vehicle-type mix in current 5-min window, time of day, supervisor availability.

Action $\mathcal{A}$: lane-type assignment {cash, EZPass} per lane and E-ZPass toll price {\$2.50, \$3.00, \$3.50}.

Decision epochs: every 15 minutes.

Reward r_t : $-\text{total_vehicle_delay} + \alpha \cdot \text{hourly_revenue}$.

RL deliverables.

- Train DQN; compare to the static lane configuration on mean vehicle delay and daily revenue during peak hours.
- Evaluate the congestion-pricing policy during an EZPass adoption ramp (35% $\rightarrow$ 90% over 6 months); report how the RL policy adapts lane assignments.

EP18 — Airport Gate Turnaround Operations

Domain: Transportation • **Difficulty:** ★★★ • **Key techniques:** Ground-resource contention, delay propagation, gate conflicts

System Overview. A regional airport with 12 gates handles 48 aircraft turns per day. Ground handling resources (cleaning crews, fuel trucks, catering vehicle) are shared

across gates. Arrival delays cascade into late departures and propagate to subsequent flight legs. The study evaluates resource levels, buffer scheduling, and remote-stand use.

Operational Description. Each turn involves a sequence of activities, some parallel. Ground resources are assigned on first-come-first-served basis from shared pools. When an aircraft arrives late and its gate is needed by the next scheduled aircraft, the late aircraft is diverted to a remote stand, adding passenger bus time. Departure delays on one rotation propagate to the next flight operated by the same aircraft.

Data.

Activity	Duration
Parking + jetway connect	5 min (deterministic)
Deboarding	Exp(15 min)
Cleaning (requires 1 crew)	Exp(20 min); parallel with catering
Catering (requires catering vehicle)	Exp(18 min); parallel with cleaning
Fueling (requires 1 fuel truck)	$\mathcal{N}(25 \text{ min}, 5 \text{ min})$; parallel
Boarding	Exp(22 min)
Pushback	Exp(8 min)

Parameter	Value
Target turnaround time	55 min
Aircraft arrival delay	$\mathcal{N}(0 \text{ min}, 18 \text{ min})$ (clipped at 0)
Gate conflict trigger	arrival delay > 20 min
Remote stand penalty	+15 min for buses
Delay propagation	50% of departure delays > 15 min carry to next rotation
Ground resources	2 cleaning crews, 2 fuel trucks, 1 catering vehicle

*Arrival delay distributions should be calibrated from the airport’s ACARS or FIDS data.
Ground-handling times can be measured from ramp video or operations logs.*

Key Performance Indicators.

- Turnaround time (mean and 90th percentile)
- On-time departure rate (target > 85%)
- Gate conflict rate
- Delay propagation factor (downstream delay minutes generated per 1-minute arrival delay)

- Ground resource utilization

Deliverables.

1. Simulate 180 days; validate on-time departure rate.
2. Resource sensitivity: add 1 cleaning crew and 1 fuel truck; report turnaround-time reduction.
3. Arrival delay impact: increase standard deviation from 18 to 30 min; report departure delay distribution.
4. Buffer scheduling: insert a 10-minute buffer between turns; report conflict reduction versus gate utilization loss.
5. Remote stand policy: compare current conflict-triggered diversion to pre-emptively scheduling 20% of turns at remote stands; evaluate both on-time rate and passenger experience cost.

Extensions.

1. Weather event (1 event per week): all arrivals delayed by +30 min for 4 hours; evaluate recovery and overnight aircraft positioning.
2. Cabin-crew rest violation: if turnaround exceeds 90 min, crew rest is violated and the next flight is delayed 60 min; quantify frequency and passenger impact.

Research Literature. Search WSC, *Transportation Research Part E*, and *Journal of Air Transport Management* for “airport turnaround simulation”, “ground handling optimization DES”, and “aircraft gate scheduling”. Identify the minimum turnaround time (MTT) benchmarks for your aircraft class and whether crew allocation has been studied as a dynamic dispatching problem in simulation.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: aircraft at each gate, ground-crew units available, remaining tasks per aircraft (fueling, catering, cleaning, loading), time to scheduled departure.

Action $\mathcal{A}$: assign the next available crew unit to a specific task at a specific aircraft.

Decision epochs: each time a crew unit becomes free.

Reward r_t : $+ \text{on-time_departure_bonus} - \alpha \cdot \text{delay_minutes} - \beta \cdot \text{crew_overtime_cost}$.

RL deliverables.

1. Train PPO; compare to priority-based heuristic (nearest aircraft by departure time) on on-time departure rate and crew overtime.
2. Evaluate under disruption (one crew unit unavailable for 90 min); report recovery time and delay propagation.

EP19 — Urban Signalized Intersection Control

Domain: Transportation • **Difficulty:** ★★ • **Key techniques:** Signalized queuing, fixed vs. adaptive timing, level of service

System Overview. A four-legged signalized intersection with exclusive left-turn lanes operates on a fixed 90-second cycle. The study optimizes phase splits, evaluates actuated (adaptive) control, and assesses the impact of a pedestrian crossing phase on vehicle delay.

Operational Description. Vehicles arrive according to a Poisson process at each approach. Queues build during red phases and discharge at the saturation flow rate during green. Overflow queues (exceeding physical storage) spill back and block upstream traffic. A pedestrian all-red phase allocates 15 seconds per cycle.

Data.

Parameter	Value
Volume N–S (through)	900 veh/hr (AM peak)
Volume E–W (through)	600 veh/hr
Left-turn fraction	15% of each approach volume
Cycle length	90 sec
Phase splits (initial)	N–S through: 35 sec; E–W through: 25 sec; N–S left: 10 sec; E–W left: 5 sec
Pedestrian all-red	15 sec per cycle
Saturation flow rate	1,800 veh/hr/lane
Start-up lost time	2 sec per phase
Physical queue storage	20 vehicles per approach lane

Traffic volumes should be collected via turning-movement counts (TMC) during the study period. Saturation flow rates can be measured directly or estimated from HCM procedures.

Key Performance Indicators.

- Average vehicle delay (sec/vehicle) by approach
- Queue length (95th percentile, vehicles)

- Level of service (HCM classification)
- Volume-to-capacity ratio per phase
- Overflow probability (queue exceeds physical storage)

Deliverables.

1. Simulate a 2-hour AM peak; validate delay against the HCM analytical formula for each approach.
2. Phase-split optimization: vary each phase split ± 10 sec while holding total cycle fixed; minimize total vehicle-hours of delay.
3. Adaptive signal: implement a detector-based signal that extends the current green by 5 sec if queue is detected; evaluate delay reduction.
4. Protected-to-permitted left: remove the exclusive left-turn phase for the minor street; evaluate delay and conflict-probability impact.
5. Pedestrian demand sensitivity: double pedestrian volume (30 sec all-red); evaluate vehicle delay increase and level-of-service change.

Extensions.

1. Connected-vehicle platoon: 30% of vehicles with V2I connectivity reduces start-up lost time from 2 sec to 0.5 sec per platoon; evaluate throughput improvement.
2. Coordinate with a second intersection 400 m away: implement a green-wave offset; evaluate arterial average delay.

Research Literature. Search WSC, *Transportation Research Part C*, and IEEE Intelligent Transportation Systems for “adaptive traffic signal control simulation”, “reinforcement learning traffic signals DES”, and “intersection control optimization”. Webster’s fixed-time formula is the canonical analytical benchmark; understand its assumptions and identify where it fails under non-uniform demand or pedestrian conflict.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: queue length per approach, current phase index, time elapsed in current phase, pedestrian demand indicator.

Action $\mathcal{A}$: next phase selection (4 options) and green duration $\{30, 45, 60, 75, 90\}$ sec.

Decision epochs: at each phase transition.

Reward r_t : $-\text{total_vehicle_delay} - \alpha \cdot \text{pedestrian_wait}$.

RL deliverables.

1. Train DQN; compare to Webster fixed-time and an actuated-control baseline on mean vehicle delay with 95 % CIs.

2. Evaluate under mixed traffic (20% non-motorized users); report the delay-equity tradeoff between vehicles and pedestrians.
 3. Situate results against published SUMO-based RL traffic benchmarks.
-

6 Computing and Information Technology

EP20 — Cloud Auto-Scaling for a Web API

Domain: Computing / IT • **Difficulty:** ★★★ • **Key techniques:** Auto-scaling policy, SLA modeling, cost–performance tradeoff, demand spikes

System Overview. A SaaS platform serves API requests from a cloud VM cluster. Traffic follows a daily sinusoidal pattern with random demand spikes. Auto-scaling provisions or removes VMs based on CPU utilization thresholds, but provisioning latency creates a window of under-capacity. The study optimizes scaling policy to satisfy a 99th-percentile latency SLA while minimizing cloud cost.

Operational Description. Each VM services requests from a shared queue with exponential service time. When aggregate CPU utilization exceeds an upper threshold for a sustained period, a new VM is provisioned after a stochastic delay. When utilization falls below a lower threshold, a VM is de-provisioned. Requests that wait more than 5 seconds time out (fail). Demand spikes arrive as a Poisson process and temporarily multiply the arrival rate.

Data.

Parameter	Value
Baseline arrival rate	1,000 req/hr (06:00–22:00); 200 req/hr (22:00–06:00)
Demand spike frequency	Pois(2)/day
Spike magnitude	$\text{LN}(5 \times \text{baseline}, 2 \times)$
Spike duration	Exp(30 min)
VM capacity	200 req/hr; Exp(0.12 sec) service
Scale-out threshold	CPU > 70% for 3 consecutive min
Scale-in threshold	CPU < 30% for 5 consecutive min
Provisioning latency	$\mathcal{N}(90 \text{ sec}, 30 \text{ sec})$
Timeout (request failure)	5 sec
VM cost	\$0.15/hr
Request failure penalty	\$0.50/failed request
SLA	p99 latency < 2 sec; failure rate < 0.1%

Traffic patterns should be extracted from API gateway access logs. VM performance parameters can be measured from load-testing runs. Cloud cost rates should be sourced from the provider's current pricing.

Key Performance Indicators.

- 99th-percentile response latency
- Request failure rate
- VM count over time (mean and peak)
- Daily cloud cost
- SLA violation events per day

Deliverables.

1. Simulate 90 days; compute SLA compliance rate and daily cost at baseline scaling policy.
2. Scaling policy search: vary CPU scale-out threshold (50%, 60%, 70%, 80%) and scale-out delay (1, 3, 5 min); plot cost vs. SLA compliance.
3. Anticipatory scaling: pre-warm 2 additional VMs between 06:00–08:00 daily; evaluate reduction in spike-induced failures.
4. Cost–SLA Pareto: for each scaling policy, identify Pareto-efficient configurations; recommend a policy with rationale.
5. Report 95% CIs for daily cost and failure rate across 30 replications under baseline and recommended policy.

Extensions.

1. Spot instance preemption: VMs have 2% hourly preemption probability with 2-minute warning; model failover (re-queue in-flight requests); evaluate cost savings vs. SLA risk.
2. Cache layer: 40% of requests served from cache (no VM needed, $\text{Exp}(0.01 \text{ sec})$ latency); evaluate VM-count and cost reduction.

Research Literature. Search *ACM SIGMETRICS*, *IEEE INFOCOM*, and *WSC* for “cloud auto-scaling reinforcement learning”, “SLA-aware scaling policy DES”, and “predictive vs. reactive auto-scaling”. Understand the gap between threshold-based reactive scaling and model-predictive approaches, and identify which workload characteristics (burstiness, seasonality) determine when RL outperforms a well-tuned threshold policy.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: current VM count, CPU utilization (trailing 3-min average), request queue length, time of day, spike-in-progress indicator.

Action $\mathcal{A}$: {scale_out, scale_in, hold}.

Decision epochs: every 60 seconds.

Reward r_t : $-\text{sla_violation_count} - \alpha \cdot \text{vm_cost_per_hour} + \beta \cdot \text{request_throughput}$.

RL deliverables.

1. Train DQN; compare to the best threshold-based policy from Deliverable 2 on SLA compliance and daily cost with 95 % CIs.
2. Evaluate robustness: test the trained agent on a workload with different spike frequency (4/day vs. 2/day training); report SLA degradation.
3. Interpret the policy: at what (VM count, utilization) state does the agent scale out earlier than the threshold rule? Quantify the anticipatory behavior.

Research Track Deliverables (Track B).

1. **Research hypothesis.** E.g., “An RL-trained scaling agent reduces SLA violations by $> 40\%$ vs. the best threshold policy under workloads with spike inter-arrival $\text{CV} > 1.5$, at equal or lower cost.”
2. **Analytical benchmark.** Model the system as $M/M/c$ with time-varying c ; derive the minimum VM count needed to meet the p99 latency SLA at steady-state load; compare to the simulation result — the gap quantifies the cost of provisioning latency and spike uncertainty.
3. **Factorial experiment.** 3×3 design: spike frequency $\{1, 2, 4\}/\text{day} \times \text{provision-}$

ing latency $\{30, 90, 180\}$ sec; compare threshold vs. anticipatory vs. RL policy across all 9 cells.

4. **Regression metamodel.** Fit a response surface for SLA violation rate as a function of spike frequency and provisioning latency; identify the boundary above which RL dominates.
5. **Generalizable conclusions.** Express the RL advantage as a function of workload burstiness and provisioning lag; relate to published cloud workload traces.

EP21 — Call Center with Skill-Based Routing

Domain: Computing / Customer Service • **Difficulty:** ★★ • **Key techniques:** Skill-based routing, abandonment, agent pools, staffing optimization

System Overview. A bank call center handles three customer segments (Retail, Mortgage, Small Business) with two agent tiers: Tier 1 generalist agents and Tier 2 specialists. Callers who wait too long abandon. The study evaluates staffing levels, skill mix, and a callback option to improve service level while controlling cost.

Operational Description. Incoming calls are classified by segment and routed to an eligible agent. Tier 1 agents handle Retail calls and, if trained, Mortgage or Small Business calls. Tier 2 specialists handle overflow from Tier 1 for complex segments. Calls that are not answered within the customer's patience time are abandoned. Agents perform after-call work before becoming available for the next call.

Data.

Parameter	Value
Total call rate (08:00–20:00)	800 calls/hr
Total call rate (20:00–08:00)	200 calls/hr
Segment mix	60% Retail, 25% Mortgage, 15% Small Business
Tier 1 agents	40
Tier 2 agents	10
Tier 1 Mortgage-trained fraction	70%
Tier 1 Small Business-trained fraction	40%
Service time (Retail)	Exp(4 min)
Service time (Mortgage)	Exp(12 min)
Service time (Small Business)	Exp(18 min)
After-call work (Tier 1/Tier 2)	Exp(1.5 min) / Exp(3 min)
Patience (Retail/SB)	Exp(5 min)
Patience (Mortgage)	Exp(8 min)
Service-level target	80% of calls answered within 20 sec

Call volumes and handle times should be sourced from the ACD (Automatic Call Distributor) system records. Abandonment patience distributions can be estimated from the ACD's queue-abandon timestamps.

Key Performance Indicators.

- Service level (% calls answered $\leq$ 20 sec) by segment
- Abandonment rate by segment
- Agent utilization by tier
- Average handle time
- Overflow rate (Tier 1 calls escalated to Tier 2)

Deliverables.

1. Simulate 90 days; validate service level and abandonment rate.
2. Staffing reduction: decrease Tier 1 from 40 to 35; add 2 Tier 2 agents; evaluate service level and total labor cost (\$18/hr Tier 1, \$24/hr Tier 2).
3. Skill-mix investment: train 20% more Tier 1 agents for Mortgage (cost: \$500 per agent); evaluate Mortgage service-level improvement.
4. Callback option: offer callback when wait $>$ 3 min; 50% of customers accept; they are re-queued 15 min later; report abandonment reduction.

5. CRN-paired confidence intervals for service level under each staffing scenario vs. baseline.

Extensions.

1. Monday morning spike: 50% above average call rate for 2 hours; evaluate service-level drop and time to recovery.
2. IVR self-service: 25% of Retail calls resolved by IVR (Exp(3 min) self-service); evaluate agent load reduction and cost saving.

Research Literature. Search *Management Science*, *Operations Research*, and WSC for “skill-based routing simulation”, “call center staffing optimization DES”, and “Erlang-A overflow model”. The Erlang-C and Erlang-A approximations are standard benchmarks; understand their assumptions (homogeneous agents, Poisson arrivals, exponential service) and identify where skill heterogeneity and overflow routing invalidate them in your model.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: queue length per segment, agents available per (tier, skill) combination, time of day.

Action $\mathcal{A}$: routing priority matrix: for each available agent, which queue to serve next; discretized as a preference ranking over segments.

Decision epochs: each time an agent becomes free.

Reward r_t : service-level_achievement $- \alpha \cdot$ abandonment $- \beta \cdot$ overflow_cost.

RL deliverables.

1. Train PPO; compare to the $c\mu$ routing rule and to the longest-queue rule on service level and abandonment rate per segment.
2. Evaluate under Monday spike (Deliverable 4): does the RL policy recover service level faster than the fixed-priority heuristic?

Research Track Deliverables (Track B).

1. **Research hypothesis.** E.g., “Erlang-C underestimates the required Tier-1 head-count by $> 15\%$ when the overflow fraction exceeds 0.20, due to service-time heterogeneity across segments.”
2. **Analytical benchmark.** Apply Erlang-C to the aggregate queue (pooling all segments); compute the predicted service level and compare to simulation; document the discrepancy attributable to skill heterogeneity and abandonment.
3. **Factorial experiment.** 3×3 : Tier-1/Tier-2 ratio $\{3:1, 4:1, 5:1\} \times$ Mortgage-trained Tier-1 fraction $\{50\%, 70\%, 90\%\}$; report service level per segment in

each cell.

4. **Policy sensitivity.** For each cell, compare skill-based routing to random routing; quantify the value of routing intelligence as a function of skill mix.
5. **Generalizable conclusions.** State the conditions under which Erlang-C becomes unreliable for SBR systems (overflow fraction, service-time CV, patience).

7 Project Management

EP22 — Software Backlog Execution with Rework

Domain: Project Management • **Difficulty:** ★★ • **Key techniques:** Developer resource pools, priority rework queues, feedback loops, time-to-release estimation

System Overview. A software product team of eight developers works through a prioritized story backlog in a continuous-flow model. Each developer is an independent server that pulls the highest-priority available story as soon as it finishes its previous task. Completed stories undergo a brief peer review occupying a second developer. Defects discovered after review re-enter the backlog as high-priority rework items, creating a feedback loop that consumes capacity and delays release. Scope creep injects new stories at each planning boundary. The study estimates the distribution of time to release and identifies the dominant schedule-risk drivers.

Operational Description. Stories are characterized by size in story points (SP); each developer's throughput converts SP into elapsed work-days with lognormal variability reflecting meetings, context-switching, and task complexity. Upon completing a story, the developer immediately starts peer review of the story (requiring one other available developer for a short fixed time) before the story is "accepted." With probability p_d , an accepted story generates a defect that is re-injected as a high-priority story of size 1–3 SP. Scope creep adds new stories to the bottom of the backlog at each sprint planning boundary (every 10 working days). The project releases when the total backlog—planned stories, items in review, and known defects—simultaneously reaches zero.

Data.

Parameter	Value
Developers	8 (each modeled as a SimPy resource with capacity 1)
Developer throughput	LN(5 SP/day, 1.5 SP/day) per developer
Initial backlog	600 story points
Story size distribution	$30\% \times 1 \text{ SP}, 25\% \times 2, 20\% \times 3, 15\% \times 5, 7\% \times 8, 3\% \times 13$
Peer-review time	$\mathcal{N}(1 \text{ hr}, 0.3 \text{ hr})$; requires one available developer (separate from the author)
Defect rate	8% of accepted stories generate one rework defect (size: Uniform 1–3 SP; priority above normal stories)
Scope creep	5% of remaining backlog added every 10 working days
Working day	8 hours; working week: 5 days
Release criterion	planned backlog + items in review + known defects = 0
Target release	80 working days (16 weeks)

Developer throughput (SP/day) should be calibrated from historical velocity in Jira or a similar tracker, adjusted for meeting overhead and context-switch cost. Defect rates and scope-creep fractions should be estimated from retrospective data. Peer-review contention is often overlooked; validate its effect by disabling it in a sensitivity run and observing the release-date shift.

Key Performance Indicators.

- Time to release (working days) — mean, 10th, and 90th percentile
- $P(\text{release} \leq 80 \text{ working days})$
- Developer utilization: fraction of time spent coding vs. reviewing
- Rework fraction: story points consumed by defect rework as a % of total
- Backlog burn-down rate (SP/day) and scope-creep injection rate

Deliverables.

1. Simulate 500 project instances; report $P(\text{release} \leq 80)$ and the expected release day.
2. Team addition: +1 developer; re-evaluate $P(\text{release} \leq 80)$ and report the marginal cost-per-day-saved at \$800/developer-day.
3. Scope-creep sensitivity: vary creep rate from 0% to 15% in steps of 5%; report release-day distribution (box plots) at each level.

4. Automated testing investment: reduces defect rate from 8% to 3% but adds 30 min of CI overhead per story; evaluate net effect on mean release day.
5. Risk attribution: use scenario analysis (one factor at a time) to decompose variance in days-to-release into contributions from throughput variability, scope creep, and defect rate; report the dominant driver.

Extensions.

1. Story dependencies: 20% of stories have a prerequisite that must be accepted before the dependent story can start; a blocked developer re-checks every 4 hours; evaluate release delay and idle-developer fraction.
2. Developer absence: each developer has a 5% probability of being absent on any given working day (illness, vacation); evaluate the release-date distribution shift.

Research Literature. Search WSC, *Empirical Software Engineering*, and *IEEE Transactions on Software Engineering* for “software project simulation DES”, “agile backlog execution model”, and “rework feedback loop Monte Carlo”. Identify what is known about the relative contribution of scope creep, defect rework, and throughput variability to schedule overrun, and whether DES has been compared to PERT-style Monte Carlo for project-risk quantification.

Advanced RL Extension (Track C). MDP formulation.

State S : remaining backlog (SP), rework fraction (rolling sprint average), developer utilization, sprints until target date.

Action $\mathcal{A}$: story-priority policy {high-SP, low-SP, rework-first} and automated-testing investment {0%, 25%, 50%} of sprint capacity.

Decision epochs: at each sprint boundary (every 10 working days).

Reward r_t : $-\text{days_to_release} - \alpha \cdot \text{testing_cost} + \beta \cdot \text{defect_reduction}$.

RL deliverables.

1. Train DQN; compare to greedy rework-first heuristic on expected release day and $P(\text{release} \leq 80)$.
 2. Analyze whether the agent invests in testing early or waits for rework rate to rise; relate to the “shift left” testing principle.
-

EP23 — Construction Project with Weather Delays and Resource Conflicts

Domain: Project Management • **Difficulty:** *** • **Key techniques:** Activity networks, resource leveling, stochastic weather, contingency buffers

System Overview. A 150,000 sq ft commercial building is planned for an 18-month schedule using a work breakdown structure of 120 activities. Weather disrupts outdoor activities; two shared cranes create resource conflicts that delay non-critical activities. The study estimates on-time completion probability and evaluates buffer and resource strategies.

Operational Description. Activities have stochastic durations and resource requirements. The critical path is derived from baseline (deterministic mean) durations. In each simulated week, weather state follows a Markov chain; bad-weather weeks suspend outdoor activities. When two activities simultaneously require a crane, critical-path activities take priority.

Data.

Representative key activities (weeks; all Lognormal):

Activity	Mean (wk)	Std (wk)	Resource
Site preparation	3	0.6	Crane
Foundation pour (×4 sections)	2	0.5	Labor
Steel erection	8	1.5	Crane
Concrete decks (×6 floors)	2	0.4	Crane
Mechanical rough-in	6	1.2	Labor
Electrical rough-in	5	1.0	Labor
Exterior cladding	4	0.8	Crane
Interior finishes	6	1.5	Labor
Commissioning	3	0.5	None

Parameter	Value
Cranes	2 (shared; critical path has priority)
Weather model	Markov chain (Good/Bad states)
Dry season (Apr–Sep): $p(\text{Good} \rightarrow \text{Bad}) = 0.15/\text{wk}$	
Wet season (Oct–Mar): $p(\text{Good} \rightarrow \text{Bad}) = 0.30/\text{wk}$	
$p(\text{Bad} \rightarrow \text{Good}) = 0.35/\text{wk}$ (both seasons)	
Indirect cost	\$10,000/week
Liquidated damages	\$25,000/week beyond 18-month deadline

Activity durations should be estimated from quantity take-offs and historical productivity data. Weather transition probabilities should be calibrated from local meteorological records.

Key Performance Indicators.

- Project completion time (weeks) — mean and distribution
- $P(\text{completion} \leq 78 \text{ weeks})$
- Expected liquidated-damages cost
- Critical-path float consumption over the project
- Crane utilization and conflict frequency

Deliverables.

1. Simulate 500 project instances; compute $P(\text{on-time})$ and expected completion time.
2. Dry-climate comparison: set $p(\text{Good} \rightarrow \text{Bad}) = 0.03/\text{wk}$ year-round; report completion-time improvement.
3. Activity buffers: add a 1-mean-duration time buffer to the five most variable activities; evaluate $P(\text{on-time})$.
4. Crane rental: add a third crane (\$500/week); evaluate conflict reduction and project-delivery improvement.
5. Critical chain: replace activity buffers with a 3-week project buffer at the end; compare $P(\text{on-time})$ to the activity-buffer policy.

Extensions.

1. Supply disruption: structural steel delivery is delayed 3 weeks with 20% probability; evaluate critical-path impact.
2. Learning curve: repetitive activities (concrete pours) improve productivity by 5% per repeat; evaluate duration reduction and schedule benefit.

Research Literature. Search WSC, *Automation in Construction*, and *ASCE Journal of Construction Engineering and Management* for “stochastic project scheduling simulation”, “construction resource leveling DES”, and “critical chain buffer management Monte Carlo”. Understand the PERT vs. Critical Chain debate and identify what simulation adds beyond standard PERT/CPM analysis for your project network.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: activities in progress (set), crane availability, critical-path float remaining, weather state (Good/Bad), weeks elapsed.

Action $\mathcal{A}$: crane assignment (which activity gets priority for the contested crane) and buffer size $\{0, 1, 2, 3\}$ weeks appended to the critical path.

Decision epochs: weekly.

Reward r_t : $-\text{project_duration} \cdot \text{indirect_cost} - \text{liquidated_damages}$.

RL deliverables.

1. Train PPO; compare to critical-chain buffer policy on $P(\text{on-time})$ and expected total cost.
2. Evaluate under climate scenario (wet-season probabilities doubled); report whether the RL policy proactively adjusts crane allocation during bad-weather streaks.

8 AI and High-Performance Computing

EP24 — GPU Cluster Job Scheduling

Domain: Computing / AI Infrastructure • **Difficulty:** *** • **Key techniques:** Gang scheduling, heterogeneous GPU resources, priority preemption, CONWIP admission, checkpoint cost

System Overview. A shared GPU cluster at a research university serves two job classes: latency-sensitive interactive experiment notebooks (small, high-priority) and long-running overnight batch training runs (large, throughput-oriented). The cluster has four compute nodes, each with eight GPUs, for 32 GPUs total. A job must acquire all its requested GPUs simultaneously on the same node (gang scheduling); a 16-GPU job must occupy two full nodes. When an interactive job has waited too long, it can preempt a running batch job, incurring a checkpoint-and-restore overhead. The study optimizes admission control and scheduling policy to meet interactive latency targets while maximizing GPU utilization and controlling cost.

Operational Description. Jobs arrive as a Poisson process and are classified on arrival. Interactive jobs request 1, 2, or 4 GPUs (equal probability); all requested GPUs must be free on a single node before the job starts. Batch jobs request either 8 GPUs (one full node) or 16 GPUs (two adjacent nodes); the scheduler waits until a sufficient contiguous GPU block is free. When an interactive job has waited more than 2 minutes and a batch job is running on a node from which enough GPUs can be recovered, the batch job is preempted: its state is checkpointed (fixed overhead), the interactive job runs to completion, and the batch job resumes from the checkpoint. A per-job failure rate causes a fraction of jobs to restart from the most recent checkpoint, if one was written within the past 30 minutes, or from scratch.

Data.	
Parameter	Value
Nodes	4; GPUs per node: 8; total GPUs: 32
Interactive arrival rate	Pois(12)/hr
Interactive GPU request	Uniform{1, 2, 4} GPUs (same node)
Interactive service time	LN(20 min, 8 min)
Batch arrival rate	Pois(4)/hr
Batch GPU request	50% single-node (8 GPUs); 50% dual-node (16 GPUs)
Batch service time	LN(4 hr, 1.5 hr)
Preemption threshold	interactive queue wait > 2 min
Checkpoint / restore overhead	10 min per preemption event
Job failure rate	2%; restart from last checkpoint (≤ 30 min old) or from scratch
GPU compute cost	\$3/GPU-hour
Fixed cluster cost	\$500/day

Job arrival rates and GPU request distributions should be estimated from cluster scheduler logs (e.g., SLURM `sacct` records). Batch service times vary substantially by model architecture and dataset size; measure from job completion records. Checkpoint overhead depends on model parameter count and storage I/O bandwidth; 10 min is representative for a 7B-parameter model on NVMe storage.

Key Performance Indicators.

- Interactive job wait time: p_{50} and p_{90} (target: $p_{90} < 5$ min)
- Batch job slowdown: actual completion time / ideal completion time if run alone on dedicated GPUs
- GPU utilization (cluster-wide, time-average)
- Batch throughput (jobs/day)
- Preemption rate and total checkpoint overhead (GPU-hours/day)
- Daily compute cost (\$)

Modeling Scope. Model job arrivals, gang-scheduling resource acquisition, service, preemption, checkpoint overhead, and job restart. Simulate 90 continuous days. Exclude network bandwidth contention, storage I/O queuing, and multi-tenant billing beyond the cost metrics listed above.

Assumptions.

- A node's GPUs are either free or committed to a single job; partial-node sharing applies only to interactive jobs requesting ≤ 4 GPUs.
- Preemption is instantaneous at the scheduler level; only the 10-min checkpoint write time counts as service overhead.
- A restarted job resumes from its last checkpoint if one was written within 30 min; otherwise it restarts from scratch (full service time redrawn).
- The cluster operates 24/7 with no scheduled maintenance windows.

Deliverables.

1. Baseline model (FCFS for both classes, no preemption): simulate 90 days; validate GPU utilization against the analytical bound derived from total offered GPU-hours per hour divided by 32.
2. Priority preemption: enable interactive preemption when wait > 2 min; compare interactive p_{90} wait time and batch slowdown to baseline with 95% CIs.
3. Backfill scheduling: allow 1- or 2-GPU interactive jobs to start in idle GPU slots on a node that is partially occupied; evaluate interactive wait-time reduction and GPU utilization gain.

4. CONWIP admission: cap admitted batch jobs to at most K concurrent jobs; search $K \in \{2, 4, 6, 8\}$; report the K that minimizes mean batch queue time subject to cluster GPU utilization $\geq 80\%$.
5. Report 95% CIs for interactive p_{90} wait time and daily cost across 30 replications under baseline and the recommended policy.

Extensions.

1. Node failure: each node has a 1% daily probability of going offline for 4 hours; jobs on that node are restarted from the last checkpoint (or from scratch if none exists); evaluate job completion rate and the value of more frequent checkpointing (every 15 min vs. every 30 min).
2. Cloud overflow: when the batch queue exceeds 10 jobs, overflow jobs are dispatched to a cloud provider ($\text{Exp}(15 \text{ min})$ provisioning delay; cost $1.5 \times$ local rate per GPU-hour); evaluate cost impact and queue-stabilizing effect versus a static CONWIP cap.

Research Literature. Search *ACM SoCC*, *USENIX OSDI/ATC*, and *WSC* for “GPU cluster scheduling reinforcement learning”, “gang scheduling simulation”, and “pre-emption checkpoint overhead DES”. Systems such as Tiresias, Gandiva, and Pollux are relevant ML-systems benchmarks; understand their scheduling assumptions and identify what your simulation model relaxes or extends.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: free GPUs per node, interactive queue depth (by GPU request size), batch queue depth, preemptions performed today.

Action $\mathcal{A}$: for each arriving batch job: {admit, defer, preempt_batch_at_node}.

Decision epochs: each job arrival event.

Reward r_t : $-\text{interactive_p90_wait} - \alpha \cdot \text{batch_slowdown} - \beta \cdot \text{checkpoint_overhead_GPU-hrs}$.

RL deliverables.

1. Train PPO; compare to FCFS, priority preemption, and CONWIP admission on interactive p90 and batch slowdown with 95 % CIs.
2. Evaluate under arrival surge (batch rate doubles for 6 hours); report recovery time for interactive jobs.
3. Compare to published Tiresias/Gandiva benchmarks; situate your RL policy on the performance frontier.

Research Track Deliverables (Track B).

1. **Research hypothesis.** E.g., “An RL admission-control agent reduces interactive p90 wait time by $> 30\%$ vs. CONWIP- K at equivalent batch throughput, by learning to preempt selectively based on job-size and queue-depth signals that a fixed- K rule ignores.”
2. **Analytical benchmark.** Model the cluster as M/G/c (one server per node); compute steady-state utilization and mean queue length; compare to simulation. Document the discrepancy from gang-scheduling and node-affinity constraints.
3. **Factorial experiment.** 2×3 : preemption threshold $\{1, 2, 3\}$ min $\times$ CONWIP cap $\{4, 6, 8\}$; report interactive p90 and batch slowdown in all 6 cells with 95 % CIs (30 replications).
4. **Pareto analysis.** Plot the (interactive p90, batch slowdown) Pareto frontier; identify whether there exists a dominant policy that improves both simultaneously vs. baseline.
5. **Generalizable conclusions.** State: “For two-class gang schedulers with load ratio $r = \lambda_I / \lambda_B$ and GPU-request imbalance δ , the optimal preemption threshold τ^* is approximately $f(r, \delta)$.”

9 Public Safety and Emergency Services

EP25 — Fire Station Location and Response Time

Domain: Public Safety • **Difficulty:** ★★★ • **Key techniques:** Spatially distributed demand, travel-time modeling, resource coverage, facility location

System Overview. A mid-sized city of 25 districts currently operates six fire stations, each with two engines. The study evaluates station locations, engine deployment levels, and a seventh station candidate to achieve the NFPA 1710 standard of 90% of calls answered within a 4-minute travel time.

Operational Description. Fire incidents arrive as a Poisson process at rates that vary by district type (industrial, residential, commercial). The nearest available engine is dispatched; travel time depends on Haversine distance and road-speed assumptions.

Multi-engine incidents require additional engines from the nearest available station. After service, engines return to their home station.

Data.

Parameter	Value
Districts	25 (geographic coordinates provided in companion file data/ep25_districts.csv)
Incident rate by district type	Industrial: 0.8/day; residential: 0.3/day; commercial: 0.5/day
Total incidents	≈ 12/day
Engine service time	Exp(45 min)
Travel speed	40 km/hr (urban); 60 km/hr (highway)
Structural fire (70%)	3 engines required
Vehicle fire (20%)	1 engine required
False alarm (10%)	1 engine; Exp(5 min) service
Current stations	6 (locations in companion file)
Engines per station	2
Response standard	≥ 90% of calls with travel time ≤ 4 min
Cost per station	\$800,000/year (fully loaded)

Companion data file: course/problem_bank/extra_capstones/data/ep25_districts.csv provides details.

Incident rates should be estimated from CAD (Computer-Aided Dispatch) records. Travel times should incorporate time-of-day traffic factors from traffic models or GPS data. Engine availability must account for concurrent incidents and preventive maintenance downtime.

Key Performance Indicators.

- Mean travel time to incident
- 90th-percentile travel time
- Coverage rate: fraction of calls with travel time ≤ 4 min
- Engine utilization by station
- Simultaneous-incident coverage failures (second engine unavailable)

Modeling Scope. Model incident arrival, dispatch (nearest available engine), travel, service, and return. Simulate a full year (365 days) to obtain stable coverage estimates. Exclude suppression effectiveness, mutual aid beyond a neighboring jurisdiction, and aerial ladder operations.

Deliverables.

1. Simulate 365 days; compute coverage rate and mean travel time by district type.
2. Station relocation: move one of the six current stations to each of three candidate locations; report coverage improvement from each move.
3. Seventh station: evaluate five candidate sites; report marginal coverage improvement and cost-effectiveness (\$/percentage-point of coverage gained).
4. Engine deployment: reduce from 2 to 1 engine at the two lowest-demand stations; evaluate the increase in simultaneous-incident coverage failures.
5. Report 95% CIs for 90th-percentile travel time under baseline and the recommended configuration (30 replications).

Extensions.

1. Major incident: an all-hands scenario with 8 simultaneous fires; model mutual aid from a neighboring jurisdiction (response time +12 min); evaluate multi-casualty coverage.
2. Dynamic relocation: when an engine is committed to an incident, a spare engine is automatically repositioned to cover the gap; evaluate coverage improvement and fuel cost.

Research Literature. Search WSC, *OR Spectrum*, and *European Journal of Operational Research* for ``fire station location simulation'', ``dynamic relocation policy emergency DES'', and ``hypercube queueing model coverage''. The hypercube queueing model (Larson 1974) is the analytical benchmark; understand its assumptions about independence of unit availability and identify where correlated simultaneous incidents invalidate it.

Advanced RL Extension (Track C). MDP formulation.

State $\mathcal{S}$: engines available per station (binary vector), incidents in progress by district, time of day, demand-rate forecast.

Action $\mathcal{A}$: reposition engine from station i to station j (or hold); taken after each dispatch.

Decision epochs: after each engine dispatch event.

Reward r_t : $\text{coverage_rate} - \alpha \cdot \text{relocation_cost} - \beta \cdot \text{response_time_violation_count}$.

RL deliverables.

1. Train PPO; compare to static (no-relocation) and an even-spread heuristic on 90th-percentile travel time and NFPA 1710 compliance.

2. Compare the RL relocation policy to hypercube-model recommendations; identify districts where the RL agent repositions more aggressively.
3. Evaluate equity: report coverage rate by district type; does the RL policy favor high-demand districts at the expense of others?

Research Track Deliverables (Track B).

1. Research hypothesis. E.g., ``The hypercube-model-derived station locations achieve $< 90\%$ of the coverage rate of simulation- optimized locations when simultaneous-incident probability exceeds 15%, due to correlated unavailability of co-located engines.''
2. Analytical benchmark. Apply the hypercube queueing model to the 6-station baseline; compute theoretical mean response time and coverage rate; compare to simulation.
3. Simulation-optimization loop. Use a 2^k factorial (station presence/absence at $k = 5$ candidate locations) to search for the coverage-maximizing configuration; report the best 1-station- addition and 1-station-relocation alternatives.
4. Coverage-equity Pareto frontier. Plot maximum coverage rate vs. maximum coverage-rate gap across district types; identify configurations where equity is not sacrificed for aggregate coverage.
5. Generalizable conclusions. State the simultaneous-incident frequency above which static location models underestimate required engine count by more than one unit.

10 Methodology Research Problems

EP26 — Simulation Metamodeling Study

Domain: Methodology • **Difficulty:** ★★★ • **Key techniques:** Kriging/GP, central composite design, cross-validation, response surface, trust-region

Overview. A discrete-event simulation model can take seconds to minutes per replication, making large-scale optimization impractical. A *metamodel*

(surrogate model) is a fast statistical approximation trained on simulation output over a designed input space. This problem asks you to build and evaluate a Gaussian process (Kriging) metamodel for a simulation of your choice, compare it to a classical linear response surface, and identify the factor ranges where the metamodel is trustworthy.

System Description. Choose any simulation from EP01--EP25, or build a new $M/M/c/K$ queue model with known analytical solution. Define four continuous factors (e.g., arrival rate λ , service rate μ , server count c , system capacity K) and one scalar response (e.g., mean queue length L_q or service level).

Representative Parameters.

- Factors: 4 continuous variables, each varied over a $\pm 30\%$ range around a nominal operating point.
- Sampling budget: 30 simulation runs for training, 20 for test (held-out validation set).
- Replication noise: 10 replications per design point; report variance.
- Central composite design (CCD): 1 center point + 2^4 factorial + 8 axial points = 25 design points (use remaining budget for augmentation).

Key Performance Indicators.

- Root mean squared prediction error (RMSPE) on the 20 test points
- R^2_{adj} of both metamodel and linear RSM
- Maximum absolute prediction error (worst-case trust)
- Kriging prediction variance at test points (uncertainty quantification)

Modeling Scope. Build both a linear response surface (second-order RSM with interaction terms) and a Gaussian process (Kriging) metamodel using a Matérn-5/2 kernel. Use leave-one-out cross-validation on the training set to tune the Kriging kernel hyperparameters. Do not use the analytical solution to tune either metamodel --- treat the simulation as the ground truth.

Deliverables.

1. Design the CCD in the normalized $[-1, +1]^4$ space; run simulations at each design point; report the response table with 95% CIs.
2. Fit the linear RSM and the Kriging metamodel; report fitted coefficients (RSM) and estimated hyperparameters (Kriging).
3. Evaluate both models on the 20 held-out test points; report RMSPE, R^2 , and max absolute error for each.

4. Map prediction variance across the factor space: identify a *trust region* where Kriging prediction variance is below a threshold (e.g., $\hat{\sigma}^2 < 0.05$ of response range²); report its coverage of the nominal operating envelope.
5. Optimization study: use the Kriging metamodel (with expected improvement criterion) to locate the minimum of the response; validate the predicted optimum by running 10 replications at that point; report the true vs. predicted response.

Extensions.

1. Sequential design: add design points at locations with highest prediction variance; evaluate how RMSPE improves per additional run.
2. Multi-fidelity: train the Kriging model on cheap coarse replications (1 rep/point) and a few expensive accurate replications (20 rep/point); compare metamodel quality to the fixed-budget baseline.

Research Literature. Search *SIAM Review*, *Technometrics*, and WSC for ``Kriging metamodel simulation'', ``response surface methodology DES'', and ``Kleijnen Gaussian process simulation optimization''. Kleijnen's textbook (*Design and Analysis of Simulation Experiments*, Springer) is the canonical reference. Identify the debate between RSM and Kriging: when does the linearity assumption of RSM fail, and how does Kriging prediction variance compare to simulation replication error? Suggested search terms: ``expected improvement simulation'', ``sequential design metamodel'', ``simulation optimization surrogate''. Key venues: WSC metamodeling tutorial track; *European Journal of Operational Research*; *ACM Transactions on Modeling and Computer Simulation*.

Advanced RL Extension (Track C). Insight. A Kriging metamodel replaces the simulation as the environment for policy search --- this is *model-based RL*. Use the trained metamodel as a differentiable surrogate to rapidly evaluate many candidate policies, then validate the best policy in the full simulation.

State $\mathcal{S}$: current factor vector (operating point).

Action $\mathcal{A}$: discrete or continuous adjustment to factor values (step size Δx).

Reward r_t : negative predicted response (minimize KPI) with an uncertainty penalty $-\kappa \hat{\sigma}(x)$ to avoid over-exploitation of the metamodel.

Decision epochs: each ``query'' to the metamodel counts as one step.

RL deliverables.

1. Train a model-based RL agent (e.g., MBPO or Dyna-Q) using the Kriging surrogate; compare to model-free PPO trained directly on the full simulation on sample efficiency (total simulation calls to reach a target KPI).
2. Evaluate metamodel-to-simulation policy transfer: does the policy found via surrogate optimization perform as predicted when run in the full simulation? Report the reality gap.
3. Ablation: replace Kriging with the linear RSM surrogate; measure how metamodel quality affects RL sample efficiency.

Research Track Deliverables (Track B).

1. Research hypothesis. E.g., ``Kriging achieves RMSPE < 0.05 of response range with 25 training runs, while a linear RSM requires > 50 runs to achieve the same accuracy for a simulation response with significant curvature.''
2. Systematic comparison. Vary training set size $n \in \{15, 25, 40\}$; report RMSPE vs. n curves for both metamodels; identify the crossover point where additional runs no longer meaningfully reduce error.
3. Trust region characterization. Report the fraction of the design space covered by the trust region as a function of n ; identify the region of factor space where linear RSM is sufficient vs. where Kriging is necessary.
4. Optimization accuracy. Compare the Kriging-predicted optimum to the analytical optimum (if your base system has one) or to a brute-force simulation grid (100+ points); report the location error and response error.
5. Generalizable conclusions. State: ``For DES response functions with curvature index $> C$ (measured by second-order RSM coefficient ratio), Kriging requires $< N$ samples to match RSM accuracy; below C , RSM is sufficient and more interpretable.''

EP27 — AI-Assisted Model Development: A Validity Study

Domain: Methodology / AI-Assisted Simulation • **Difficulty:** ★★★ • **Key techniques:** Face validity, output comparison, V&V, AI copilot workflow, KS test

Overview. This problem is the empirical core of the course's central thesis: *simulation without a black box, in the AI era*. You will build the same queuing simulation model twice --- once using an AI copilot following a structured prompt workflow (the EP01 blueprint), and once from scratch using only the problem statement and your own judgment. Then you conduct a rigorous validity study comparing the two models. The result is publishable at the WSC practice/education track.

System Description. Choose a moderately complex system: a call center, outpatient clinic, or small distribution center. The system must be specifiable by a 1-page description with clear KPIs and representative parameters (you may use any EP01--EP25 problem statement as the specification).

Representative Parameters. The parameters are *identical for both models* --- the experiment controls for system specification; the variable is the development method.

- AI-assisted model (Model A): follow the EP01 AI Copilot Blueprint exactly --- conceptual model prompt, SimPy scaffold prompt, validation prompt, scenario analysis prompt.
- Independent model (Model I): build from the problem statement without any AI assistance; design your own event logic, classes, and data structures.
- Time budget: cap development time for each model to ≤ 8 hours; log effort in 30-minute increments.
- Seed all random number streams identically for a fair output comparison.

Key Performance Indicators.

- Structural validity: fraction of conceptual-model components correctly captured (use the EP01 conceptual-model checklist)
- Parameter validity: mean absolute percentage deviation in model parameters (service rates, routing probabilities) between the two models
- Output validity: KS statistic on 30-replication output distributions (mean queue length, mean wait, throughput) between Model A and Model I, and between each model and any available analytical benchmark
- Development efficiency: hours-to-first-valid-run, lines of code, defect count (bugs that caused incorrect output)

Modeling Scope. Build two complete, independently executable SimPy models for the same system. Both must pass at least the following V&V checks: degenerate-

input test (zero arrivals $\rightarrow$ empty system), extreme-load test ($\rho \rightarrow 1$: queue grows unboundedly), and replication stability (mean converges as $n_{\text{reps}} \rightarrow 30$). Document every bug encountered during development of each model.

Deliverables.

1. Structural comparison. List all conceptual-model components (entities, resources, queues, decision points, performance metrics); for each, indicate whether Model A and Model I both capture it, only one captures it, or it is captured differently. Compute the structural agreement percentage.
2. Parameter comparison. Tabulate all numerical parameters in both models; compute mean absolute percentage deviation; flag any parameter where the two models differ by $> 5\%$.
3. Output comparison. Run 30 replications of each model with the same seeds; apply the KS test to the KPI distributions; report D_{KS} and p -value for each KPI. If an analytical solution exists (e.g., M/M/c formula), include it as a third reference.
4. Development audit. Report hours-to-first-valid-run, bugs-per-100-LOC, and the nature of each bug (logic error, off-by-one, simulation termination, statistical reporting). Classify bugs by whether the AI prompt or the developer's design choices introduced them.
5. Generalizability analysis. Based on your findings, state a testable proposition: under what conditions (system complexity, developer experience, prompt quality) does AI-assisted development produce a structurally equivalent but different-parameter model vs. a structurally divergent model?

Extensions.

1. Replicate the study with a second developer building Model I independently; report inter-developer variability and ask: is AI-assisted development *more consistent* than human-independent development?
2. Evaluate a third model (Model B) built by asking the AI to validate Model I: give Model I's code to the AI and ask it to identify errors and inconsistencies with the problem statement. Does AI review close the gap between Model A and Model I?

Research Literature. Search WSC practice and education tracks, *Empirical Software Engineering*, and *ACM TOSEM* for ``AI-assisted software development empirical study'', ``LLM code generation quality'', ``simulation model validity

AI'', and ``copilot productivity measurement''. Key reference contexts: GitHub Copilot empirical studies (Imai 2022; Vaithilingam et al. 2022); WSC education track papers on simulation pedagogy; INFORMS *Service Science* on simulation-based decision support quality. Understand: what is currently known about AI-generated code correctness? How do published studies define and measure ``code quality''? Identify the gap: *no published study has applied simulation V&V methodology to AI-assisted DES model development* --- this is the contribution. Key venues: WSC practice/education track; *Winter Simulation Conference* best paper award in education; *Simulation: Transactions of the Society for Modeling and Simulation International*.

Advanced RL Extension (Track C). Insight. This problem itself becomes a simulation-as-environment study when you ask: *can an RL agent learn to select the best AI prompt sequence to minimize development defects?*

State S : current model code (feature vector: LOC, component completeness, test-pass count), prompt history (sequence of prompt types used).

Action A : select the next prompt type from: {conceptual, scaffold, validate, debug, scenario}

Decision epochs: each prompt submission.

Reward r_t : $-\text{bugs_introduced} + \text{components_completed} - \alpha \cdot \text{prompts_used}$.

RL deliverables.

1. Instrument the EP01 AI workflow as a simulator (replay transcripts from multiple students; model prompt $\rightarrow$ code outcome stochastically).
2. Train a DQN agent to sequence prompts; compare to the fixed EP01 Blueprint sequence on bugs-per-100-LOC and hours-to-valid-model.
3. Report the most common optimal prompt sequence: does the agent converge on the Blueprint order, or does it discover a better sequence?

Research Track Deliverables (Track B).

1. Research hypothesis. E.g., ``AI-assisted development achieves structural agreement $\geq 90\%$ with independent development and reduces hours-to-first-valid-run by $\geq 50\%$, but introduces a higher fraction of statistical-reporting bugs (KPI miscalculation) vs. logic or structure bugs.''
2. Controlled experiment. Recruit 2--4 additional developers to build Model I independently; report the distribution of structural agreement across human developers vs. the AI-assisted model.
3. Bug taxonomy. Classify all bugs from all models into: (a) simulation logic, (b) statistical reporting, (c) parameter misspecification, (d)

structural omission. Report the distribution by development method; test whether the distributions differ significantly using Fisher's exact test.

4. Output consequence analysis. For each bug found, rerun the model with and without the fix; report the change in KPIs; classify bugs as *consequential* (KPI change $> 5\%$) or *cosmetic*. Report the fraction of consequential bugs by development method.
5. Generalizable conclusions. Frame findings as actionable guidance: ``When using AI-assisted DES development, invest X minutes in prompt Y to reduce type- Z bugs by a factor of ...'' Target submission: WSC practice/education track or *Simulation* journal.

Companion files. The course/problem_{bank}/extra_{capstones}/data/directory contains representative data (e.g., *shoprouting* table; *EP11* order-zone assignment distribution). Instructors may substitute real data.

Solutions. Full solutions---including the exact LLM prompts used to generate starter code, annotated Python models using the *simdes* package, output-analysis notebooks, and a model report template---are distributed to confirmed instructors only, as part of the instructor courseware package. Contact the author to request it for course adoption.